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Design and implementation of Mirror Therapy using VR Exergame with a bimanual robotic exoskeleton

by

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Declaration

I declare that wherever specific reference is made to the work of others, the contents of this Thesis are original and have not been submitted in whole or in part for consideration for any other degree or qualification in this, or any other university. This Thesis is my own work and contains nothing but the outcome of work done in collaboration with Aries Lab (Heidelberg University), except as specified in the text and Acknowledgments. This thesis contains fewer than 65,000 words including appendices, references, tables, and equations, and has fewer than 150 figures.

Vivek Vijaykumar Ingle
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Abstract

The purpose of this study is to conduct a preliminary stroke rehabilitation protocol by Robotic Mirror Therapy on Healthy subjects by playing a space shooting game with their dominant hand. The study aims to evaluate how the game difficulty affects the User performance and test the Mirror Haptic Feedback (mirror modality) compared to the Haptic feedback (Normal Modality). The equipment used in this project were Alex a 6 Dof Exoskeleton and a 3D Divisor which is Oculus Rift S a virtual reality Headset and the Software that was used to design and implement the Space shooting game originally created by Daniel Stringer was done in Unity 3D and the program used to develop it was C# which was done in Visual Studio. Beyond this, the thesis starts with the introduction about the topics in more detail and the results will be based on the data that was received while performing the experiments and then the data was analyzed in the software called Matlab in order to check the Performance.

Nomenclature

SHM Shared Memory

DoF Degree of Freedom

FPS Frame per Second

UI User Interface

VR Virtual Environment

ROM Range of Motion

SS Safe Space

RQ Research Question

MF Maximum Failure

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Part I

Section One

Chapter 1

Introduction

1.1 Overview

In recent years Stroke has become the leading cause of long-term disability and also death in the developed countries (1). This appears to be an important topic to study because it has a great effect on the social and economic levels as there are around 9 million Stroke survivors and every year the death rate is evaluated at around 4.5 million. The probability of increased cases in the coming years would go around 30 percent or more compared to the last three decades.(2). In such cases, there is wide popularity use of robotic devices for stroke Rehabilitation purposes in order to improve motor learning by combining the impedance and control strategies(3) Among other techniques, classic Mirror Therapy is widely popular and effective for such rehabilitation purposes but is limited to passing incomplete information to the affected hemisphere. So according to Prof. Dr. Lorenzo Masia, there can be an approach of new therapy which is Robotic Mirror Therapy after stroke referred to as ROBMIT (<https://www.ekfs.de/en/scientific-funding/currently-funded-projects/robmit-robotic-mirrored-therapy-after-stroke>).

This rehabilitation promotes the movement of the affected arm and thus among other robotic devices, the ALEx-RS robotic device (<http://www.wearable-robotics.com/kinetek/>) which is the best fit for this study (see 1.4). This device is a Bimanual robotic Exoskeleton with 6 DoF intended for the rehabilitation of the upper limb function. and using this robotic device we perform a Bimanual task by mirroring both the affected and the non-affected arm in order to improve motor learning of the subject as well as the assisted movements with the sensation of touch during the task performed in the VR environment.

1.2 Literature Review

The foundation of this study is based on the previous research. One of this research was the Starting point of my thesis. "Bimanual Motor Strategies and Handedness Role During Human-Exoskeleton Haptic Interaction" is a study based on bimanual object manipulation which is one of the aspects of motor control. It is the involvement of haptic control and the motor commands which are managed by the nervous system. This study goes into the exploration of how the hands interact with the manipulation of an object via two impedance-controlled exoskeletons to simulate in the virtual environment which also includes haptics that give you the sensation of touching the object. The results of this study were that the haptics can be affected by the bimanual strategies which rely on the direction of the movement (4). In the Human body there are various senses and one of them is the sense of touch which is described as Haptics which can be used to interact with various technologies related to computer interaction and applications. There are many cases where in this technological world people are more interested in doing activities which can be done from home rather than going out. In short, they prefer indoor activities rather than outdoor and this has led to minimizing physical exercise and unhealthy body movements which leads to unhealthy joints, weaker bones, and less physical strength. To deal with this issue this study is based on the Design and development of a new immersive platform based on Exergames and Exosuits with assistive feedback. Another study which was related to my Topic was "Mirror-Based Robotic Therapy for Ankle Recovery with a Serious Game: A Case Study with a Neurological Patient". In this study Mirror-based robotic therapy is used as a rehabilitation where the visual and motor feedback policy was based on a serious game in conjunction with Mirror Therapy where the result was positive with the range of motion for the active ankle plantarflexion and also muscle activation as this study was conducted for the ankle recovery for lower limbs (5). With respect to both studies, This particular study aims to fill this gap by examining the effects of game difficulty on performance and physiological responses by testing and comparing two Modalities which is Haptic Feedback and Mirror Haptic Feedback Modality. I started working on this project in order to reach the target of Design and Implementation of Robotic Mirror Therapy with a Bimanual Robotic Exoskeleton using VR Exergame which is a space shooter VR game originally created by Daniel Stringer (<https://www.udemy.com/user/daniel-stringer-4/>). combining all these modules, especially for upper limb neuromuscular rehabilitation a new approach can be found for implementation and beneficial compared to other traditional techniques in addition, we can find the answer to whether the difficulty levels of the game will be played in VR affect the user performance

with the exoskeleton and dose the Mirror Haptic feedback results in a better performance compared to the Haptic Feedback.

1.3 Haptic Feedback (Normal Modality)

Haptics is basically the feeling of touch when you interact with the device and in this case, it is a sensory feeling where you interact with the gun to grab in the 3D world and you can feel it through the end effectors of the Exoskeleton in the real world.

In this modality, both of your hands can move independently of each other to do a Bimanual Task. when you try to grab an object, you will need to apply enough force to have control over the object so that it does not fall or break into pieces as shown in the figure 1.2.

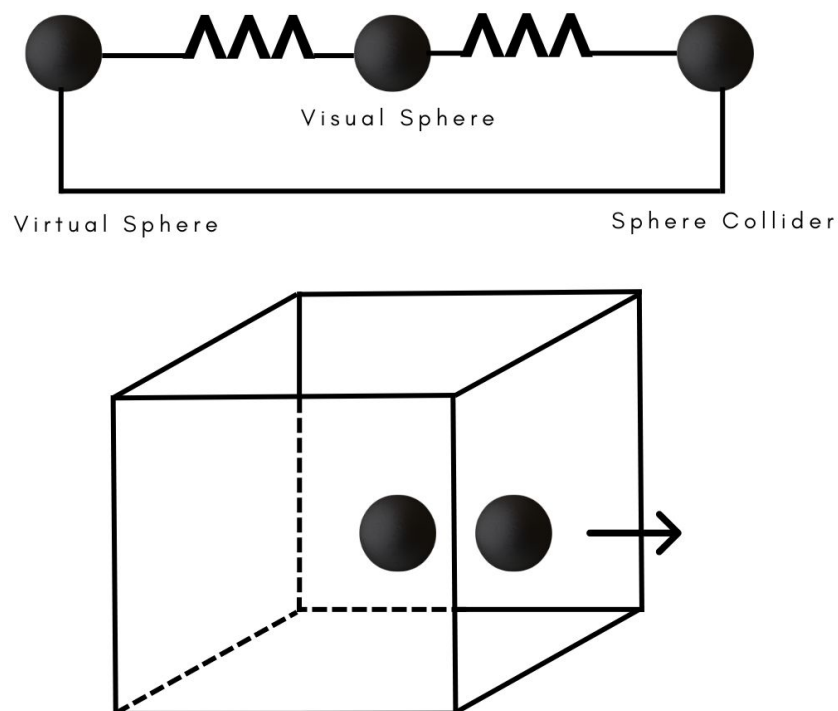


Figure 1.1 Representation of how the Haptic Feedback works

When you apply a force we use the easy relationship of the spring that is

$$F = -K(\Delta X) - C(\Delta V) \quad (1.1)$$

where K is stiffness and C is dumping as it is related to displacement and V is the velocity.

$$F = K.\Delta X \quad (1.2)$$

The Control Architecture Scheme shown in 1.2 is as follows. (A) The user wearing the exoskeleton ALEx-RS. (B) High-level controller (Virtual Haptic Unit - VHU): a visual representation of the interactive scenario. Here is computed the “physical-based” response to the user’s movements in the virtual environment, generated by (C) Impedance Controller module: this block computes the haptic feedback. The force generation is calculated by means of elastic (K) and viscous (C) components, which are proportional to the distance between the End Effector spheres ($\dot{x}_{EE} : \dot{x}_{EEI}$ or $\dot{x}_{EEr}$) and the God-Objects ($\dot{x}_{GO} : \dot{x}_{GOI}$ or $\dot{x}_{GO r}$). (D) Torque Computation module: here the force values at each EE (F_{EEr}, F_{EEI}) are then sent back to the device, and then converted in torques at the joints (τ) of each exoskeleton, by means of the transpose of the Jacobian (J^T)). using Unity 3D (6)

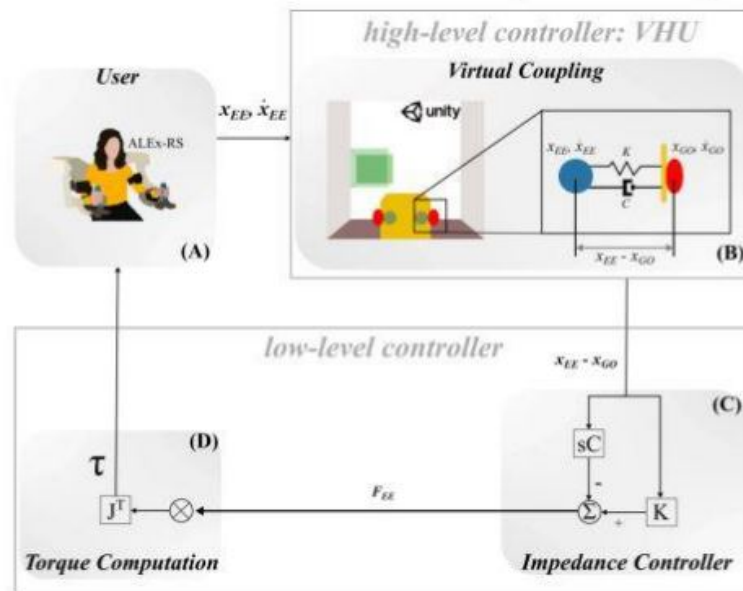


Figure 1.2 Representation of the virtual coupling (The following image is taken from <https://ieeexplore.ieee.org/document/10123352>)

1.4 Mirror Haptic Feedback (Mirror Modality)

The classical mirror therapy will tell the patient to always see your healthy hand and the other one has a mirror in front of it 1.3. So, you tell the patient to do something symmetric. so they are supposed to think of doing the same thing with the non-dominant hand as they are doing with the dominant hand to plasticity also for the non-dominant hand at some point, it starts to move more and more, and eventually, you will hope to do the same again symmetric But the problem is that you don't get any proprioceptive feedback because your hand does not move So with this in conjunction of robotic mirror therapy you provide proprioceptive feedback at some point while you imagine your hand to be moved. So the task we want to give the subject is that they also plan actively on the impaired hand to help and ensure the haptic feedback. That's the critical part and the trick and motivation of this Project If you tell the subject to do this they will do it knowing that they just have to move the right hand to do the process and not think about the left hand as It helps you see the left hand move and to see it yourself it moves and it contributes.

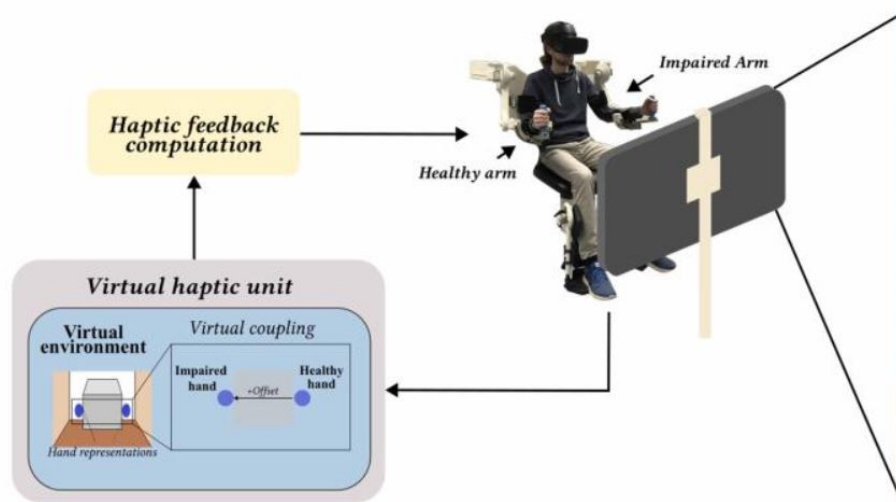


Figure 1.3 Mirror haptic diagram (The following image is taken from <https://www.ekfs.de/en/scientific-funding/currently-funded-projects/robmit-robotic-mirrored-therapy-after-stroke>)

In this modality, both hands work together to complete a Bimanual task. When you try to grab an object 1.4 which is a gun in the virtual environment by which we shoot the asteroids here both your hands are fixed on it and once it is grabbed then your dominant hand controls the movements while the non-dominant hand mimics its movements thus it's called Robotic mirror therapy as both the hands of the exoskeleton are mirrored, and one follows the other. In this scenario, we can choose which hand to be a leading one depending on the preference of the subject as they can make their own choice to shoot the spawning asteroids.



Figure 1.4 The Virtual representation of the Shooting Game in Mirror Modality

1.5 Research Question and its Approach

The research question gives you an objective and a way to solve an existing problem which is the highlight point in any study (7). In my study, there were many questions and aspects but the main were found on the basis of previous studies which were already done on this particular exoskeleton. There were two questions that came out in this research with the approach of comparing both the modalities and seeing which one is better in terms of the performance of the user. so the questions were as follows.

1. **Does the difficulty level in the VR exergame impact user performance with the exoskeleton ?.**
2. **Does mirror haptic feedback result in improved performance compared to Haptic feedback?.**

As mentioned before 1.2 This study is based on the previous research which was to create a sense of touch while performing the rehabilitation task and after successfully achieving the haptic feedback the goal was to create a mirroring effect which was to perform the same or similar task with one leading hand and the other following it. To answer the above questions first we need to understand the background for this project and the time I spent in Aries Lab I found a few aspects with the help of my supervisors that especially when you design an Exergame and configure it with the Exoskeleton we come across many challenges and a big challenge was that both devices have their own specification and limitations and keeping it into the consideration we have to build the system in such a way that both the devices work coherently where there is a specific version of software and plugins to be used and only then the whole system works without any errors, one of the main challenges which we faced is explained in 3.6.2 and then comes the approach to this study which was to compare both the modalities and make the game more effective by putting three levels where the difficulty of performing the task increases and we can determine the desired results.

Chapter 2

Background

2.1 Exoskeletons

An Exoskeleton is a wearable Mechanical structure just like a human body with joints. It is mostly used in Rehabilitation and Virtual Reality simulations which can be useful for both the disabled and the healthy people (8). In my case, It is a device that assists the movements of paired and impaired hands further leading to rehabilitation purposes. There are two types of Exoskeleton one is **Passive Exoskeleton** and the other is **Active Exoskeleton**. The active exoskeleton is dependent on a DC motor or hydraulics whereas the Passive exoskeleton is dependent on the mechanical components such as springs or elastic. An active one requires non-stop power input whereas on the other hand, the passive one is limited to the users movement (9) .

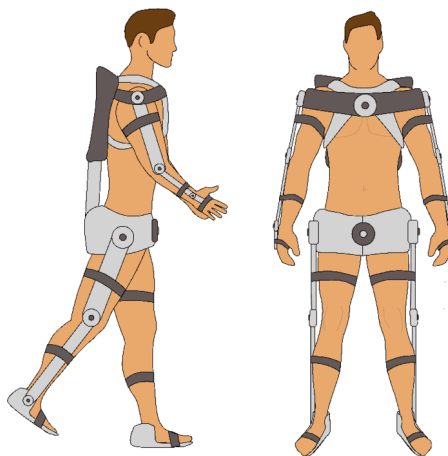


Figure 2.1 Exoskeleton Example
(10)

2.1.1 Alex-RS

The Exoskeleton which i used in this project is 2.2 a Bimanual ALEx-RS device which is for the Upper limb (11). It is a 6 DoF Exoskeleton whose four DoFs are sensorized and actuated joints (the shoulder abduction-adduction, SH-Abd-Ad, rotation, SH-Rot, and flexion-extension, SH-Flx-Ext, and the elbow flexion-extension, EL-Flx-Ext) and two sensorized and passive joints (the forearm pronosupination, FO-Pro-Sup, and the wrist flexion-extension, WR-Flx-Ext).

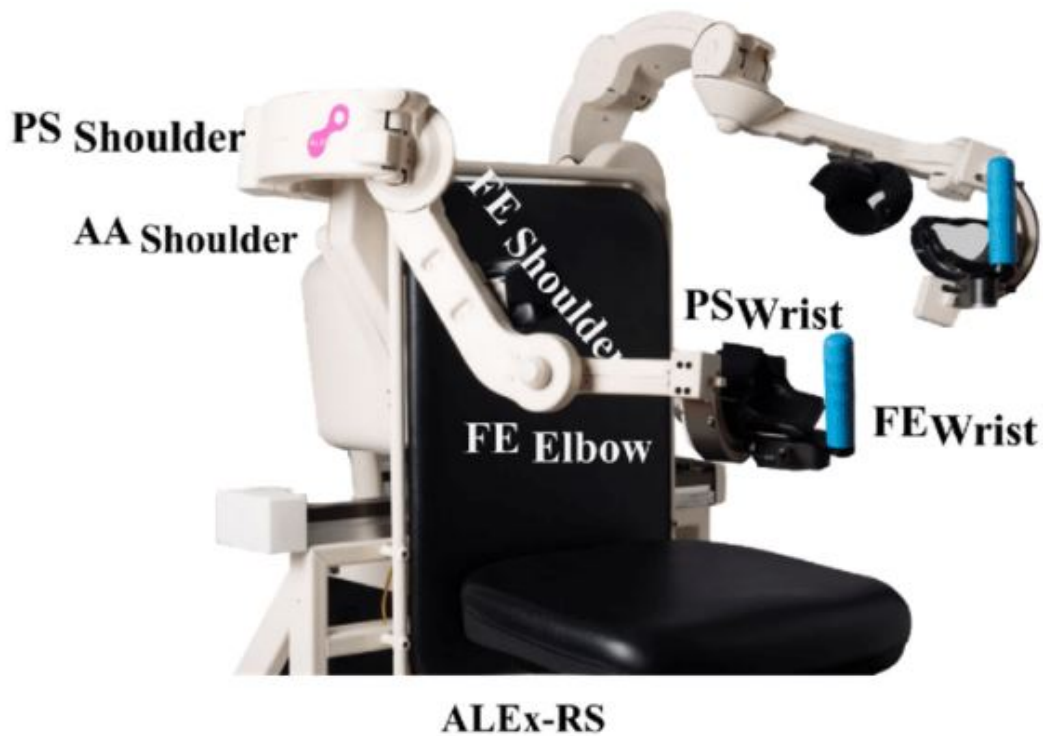


Figure 2.2 ALEx-RS (This image is taken from <http://www.wearable-robotics.com/kinetek/products/alex/>)

Alex has been designed for the movements of the human arms for rehabilitation. It can be easily worn and just needs a couple of minutes to set it up as it is a wearable device. It can be controlled with the application whether it be the force or the movements of the arms. The maximum continuous force is around 50 N and the maximum peak force is 100 N. Maximum continuous joint torque is 40 Nm and the maximum peak torque is 80 Nm (<http://www.wearablerobotics.com/kinetek/products/alex/features/>)

2.2 Stroke Rehabilitation and Exergaming in Robotics

Stroke is a medical term when occurs the blood flow in the brain is blocked due to the immense blood flowing than usual. the most general or common is **Ischemic stroke** around 80 percent which is a bit minor than the one called **Hemorrhagic** which is around 20 percent. The symptoms of stroke include sudden dizziness, and numbness of the face, leg, or arm, and is one of the leading causes of death after cancer and other heart diseases (12). As this is a serious medical condition there are many therapies and stroke rehabilitation plays a major role in improving motor learning using the Exoskeleton 2.3 as it is the main objective. Exergame is a type of exercise, with wellness or rehabilitation purposes, that promotes exercise in a fun and interactive manner. The development of such an application on Exosuits with variable motor assistance could be innovative and effective solutions to improve standard therapy. Exergaming in Robotics plays a major part in this process as this small exercise leads to the betterment of the impaired hand and recovery in this case. Exer comes from the word exercise and gaming is the video games that we play for entertainment purposes combining this is called Exergaming (13) and with the field of Robotics, it is more convenient to complete these therapies as this promotes the involvement and repetitive tasks for the subject which improves there motor learning and studies have proven that it gives the outcome in a positive way and can help in the functionality of the impaired part eventually. and in the Robotics field, there are many ways to do it but in this thesis, the main goal is on the upper limb of the body and thus we use Ales-Rs exoskeleton to carry out this exergaming in conjunction with virtual reality and a space shooting game.



Figure 2.3 User performing the bimanual task (This image is taken from <http://www.wearable-robotics.com/kinetek/products/alex/features/>)

VR Rehabilitation games come under two types based on which source the game comes from for example, one is a commercial game whereas the other is a custom-designed game commercial games are targeted towards young people and have some limitations when it comes to the rehabilitation as their difficulty level is a bit high and can impact the scores of the subject and can make them feel frustrated so that's where the custom-designed games play an important role which can be adjusted according to the preference of the subject and is easy to play and can be helpful in the data analysis as well (14).

2.3 working of the Haptic Feedback in Rehabilitation

Haptics in this case is a generation of dynamic interactions in VR. **How does it work?** You have an object, and you have the position of two hands and then once you take your hands and grab the object you will perceive the weight and inertia of the object. There is a graphic engine, we have virtual reality. The interaction between these three different bodies is calculated from the dynamic engine of the system itself. We extrapolate the information of the collision between the objects, and we send all the information to the low-level control of the system in such a way that the system then mobilizes these two arms and provides the sensation of force. We study the way of generating mirror therapy using this kind of system. We create an environment in which you can have one arm impaired and the other arm unimpaired and when you touch the object the two arms are linked with each other and can feel the haptics. The unimpaired arm drives the object in some kind of task which in my case is a space shooter exergame. And the arm which is impaired is locked to the object where we can wear the VR device and then we can simulate the gun. It is a spongy object so we can perceive the stiffness as there is spring and weight and is used for assistive feedback.

2.3.1 Spheres

There are a total of 6 Spheres where 2 of them are paired and 2 of them are hands. Hands are on the position from Alex's hands and every frame from the game. So Just say OK we put the sphere in the position of the hand (Other spheres are the two which together Follow their respective hand) two spheres from the left hand and two from the right hand follow the position of Alex in the virtual space. We do this because we need the haptic feedback. (For instance, If we go inside with the left hand in the gun other two spheres try to follow inside the cube but they have a collision and the hand does not go through) And it does not reach the cube so there is a force present towards the cube. So they take just the force and

put it on your hand in another direction. (so the sphere tries to come in from the left. they press the hand to the left and it feels like you are pressing the gun). The Spheres are called by their names. Two hands are **mano sinistra** and **mano destra** These two hands are the real positions of the hands which is the end effector of Alex. There are virtual spares that try to follow the hands and they are called as VL and VR which are virtual left and virtual right. Then there are the visual spheres and they are at the same time as the virtual spheres as it is for the compression of the gun. The previous students who worked on this project separated the visual and virtual spheres. The visual sphere is an Extra sphere that is kept a little bit nearer the cube so it looks like pressing against the gun and then we feel the Haptic feedback.

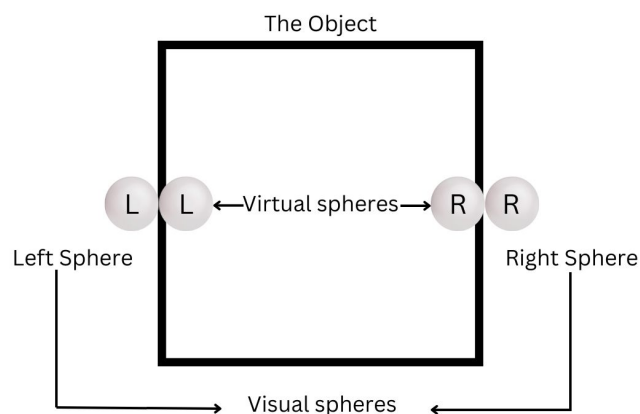


Figure 2.4 Visual and Virtual spheres

2.3.2 Forces

Force consists of two different sources.

1. Distance between the hand and the sphere where the virtual sphere tries to follow the hand which is the distance in the position to reach the position of hands. The hand is in there but there is a force of the virtual sphere against the gun.

2. Velocity difference between the two. The difference in the velocity means if the hand is faster, the virtual blue sphere follows faster.

```
float k_right = 23, k_left = 23, k_right_init = 15, k_right_min = 8;
float c = 0.5f;
```

These are the Values that are used to calculate the forces. Parameters float K right is equal to 23, K left is equal to 23, (the higher the number higher is the force). Multiplied by Force we compute the force. Which is Multiplied by K right or k left where K is a parameter to write position and force. If I put it very high, I need to reach the same position but C is multiplied by Force in different ways as the force consists of two parts distance and velocity. C is to reach the same velocity. If c is very high you say you want to have the same velocity but if you don't have the same position. for example **k = 0 and c = 1**, the sphere tries to reach the same velocity but not the same position. Then there are Reset points which are the two planes to reset the table or game. For the safe space, there is a saturation limit of 5 newtons and if the force applied is higher then the cube will turn its color in red to warn the subject so they can apply the moderate force that is necessary.

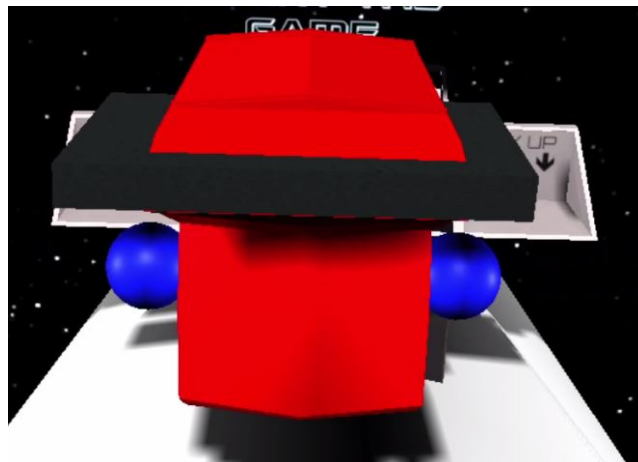


Figure 2.5 Colour change warning.

2.4 Virtual Reality

Virtual Reality is a technology that gives you a digital experience of a 3D world and gives you the feeling as if you are in some real-world environment. In technical words, it is a

Human-Computer Interface that simulates a real-world environment where the user can move freely in a safe space and interact with the objects around (15). This technology is used in many different fields, especially gaming, and entertainment, and has raised treatment and therapy approaches in stroke rehabilitation on a high note. Virtual Reality in my case is for rehabilitation purposes where the user performs the assisted tasks and you get the result out of it which further decides the outcome of whether the task is effective or not effective for the user 2.6 as the goal is to improve the performance of the user. In order to get the experience of Virtual reality we need to wear an external device which is called a 3D device or a VR headset it is connected through the Unity 3D software and the necessary plugins are installed for smooth working. There are various devices but the one that we used in this project is **Oculus Rift S**. promoting the use of VR technology for rehabilitation therapies can be very useful as it has many types of sensory input as this technology contains repeating the specific tasks and receiving continuous feedback which improves motor learning (16).



Figure 2.6 User experiencing VR environment 2.5

Oculus Rift S

Oculus Rift S is a VR headset which is developed by two platforms that is Lenovo Technology and Oculus VR in 2019. It has two controllers which are the Oculus touch controllers https://en.wikipedia.org/wiki/Oculus_Rift_S. Well in this project the controllers are not used much but just for the calibration purpose to set and create the safe space where the user will perform the rehabilitation tasks.



Figure 2.7 Oculus RIFT S

2.5 Robotic Mirror Therapy

Robotic Mirror Therapy is a way to deal with the paretic arm in order to improve its muscle movements which are affected after suffering from stroke with the help of the movements of the non-paretic arm which leads the tasks into the therapy. Here the Bimanual Robotic Exoskeleton (see figure 3.2) allows the moments of both the arms to interact and this is when the paretic arm follows the actions of non-paretic arm (17). Unlike the classic Mirror therapy where the Mirror is placed in front of the Subject and then the mirror therapy is performed or done to improve their motor learning, Robotic Mirror Therapy is a bit different but is a method adopted from it in order to reinforce (18). In classical mirror therapy, there is an activation of neurons but there is no motion of the hand that is affected. In this case, we have a motion. This therapy gives the subject the illusion of the paretic arm to be in movement and this movement helps them to improve their motor learning. The particularity of this therapy is that we are able to simulate bimanual interaction. The physics of the object that is transferred in the two arms is recognized as This is robotic assistance. If the person is not able to move, the system drives you in the proximity of the correct position and then you have to adjust accordingly as we can increase the velocity.

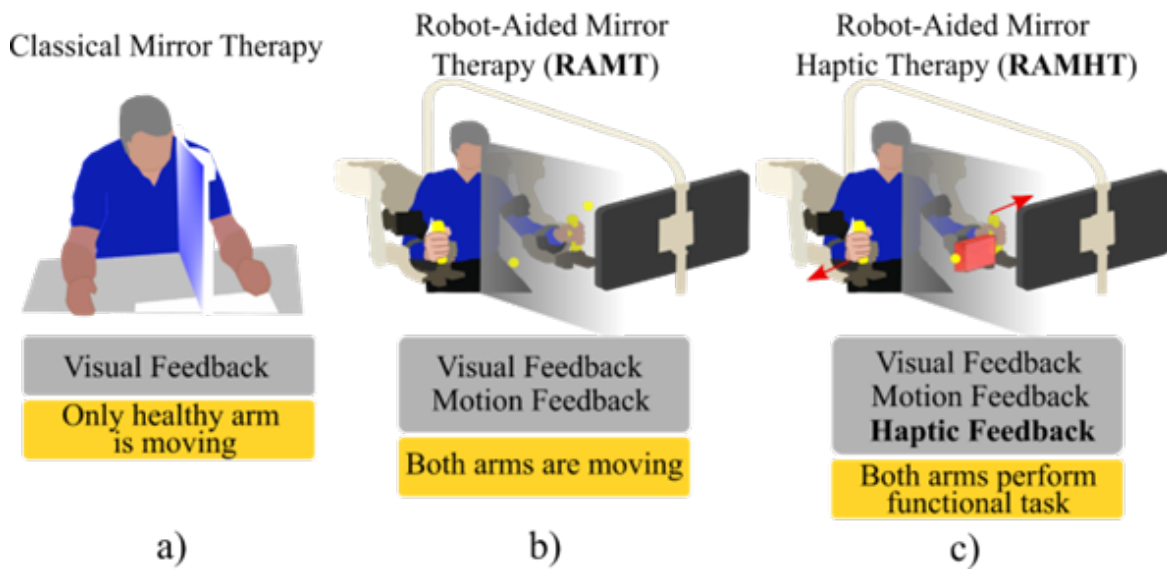


Figure 2.8 Difference between Classical and Robotic Mirror RAMT and RAHMT

2.6 Hardware and Software Setup

Setting up the hardware and the software and their calibration is not complex but tedious and that we can understand it in the steps listed below. There are totally two pieces of equipment that is the hardware used for this project the one is the Alex-RS Exoskeleton with the 6 DoF and the other is Oculus Rift S VR headset

To start and use the Alex application on PC

1. Close all windows except power 312
2. Go to the External folder
3. Double-click on Alex Translocce.
4. then double click on dll
5. Then press the button launcia tutto which will launch the program.
6. when the program starts and you can see the following page it should display as connected.
7. Do not touch other features in the UI as it is very sensitive so always keep on wearing mode as shown in the figure 2.9 while working.

8. Put the rehab mode only when the program is running and when the subject is in the Exoskeleton.

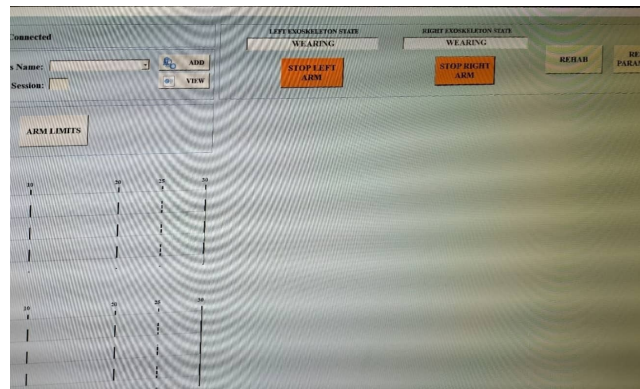


Figure 2.9 Wearing Mode

Mirror modality in the system

Mirror modality in the system is inbuilt but is limited to a few movements 3.6.2. but we can access it through the following three steps.

1. Go to the section Patient name.
2. Then go to the section treatment.
3. and then select Mirror.

To open the shell ALEX 32

The SHM which is a shared memory is where you get the real-time data of the exoskeleton to view it we have to follow the below steps.

1. SHIFT + ALT + F2 will open the SHM.
2. APP DATA IN reads the data of the exoskeleton in real-time.
3. APP DATA OUT writes the data which means is an output of the movements and the forces of the exoskeleton in real time,

4. dx represents the right hand and the dy represents the left hand of the Exoskeleton.

dx	dy
right	left

The joints and movements of 6 DoF's

Table 2.1 Degree of Freedom

Joints	Movements
0	ABDUCTION, ADDUCTION, SHOULDER
1	PRONATION, SHOULDER
2	NO
3	NO
4	FLEXION, EXTENSION, SHOULDER
5	FLEXION, EXTENSION, ELBOW

Setting up the VR Headset

To set up the Oculus Rift S we follow the steps below.

1. Install the Oculus app
2. Plug in the headset
3. Set up the guardian which is the safe space to carry out the activities.

Software used

1. **Unity 3D:** Unity Software is a Game engine and designing software that can be used to design in various platforms in my case for Virtual Reality 2.10. It is more famous in the gaming industry where we can create 2D and 3D games and use them for various interactive simulation experiences. We use this software version 2020.3.12f1 in this study to create a virtual scenario where the user will be doing the activity wearing the VR headset and experiencing a whole new world. In this thesis, it is used for the implementation of virtual reality and to design an experimental protocol. In addition, we will also use the programming language c# to program and control the 6 DoFs robotic exoskeleton Alex-RS.

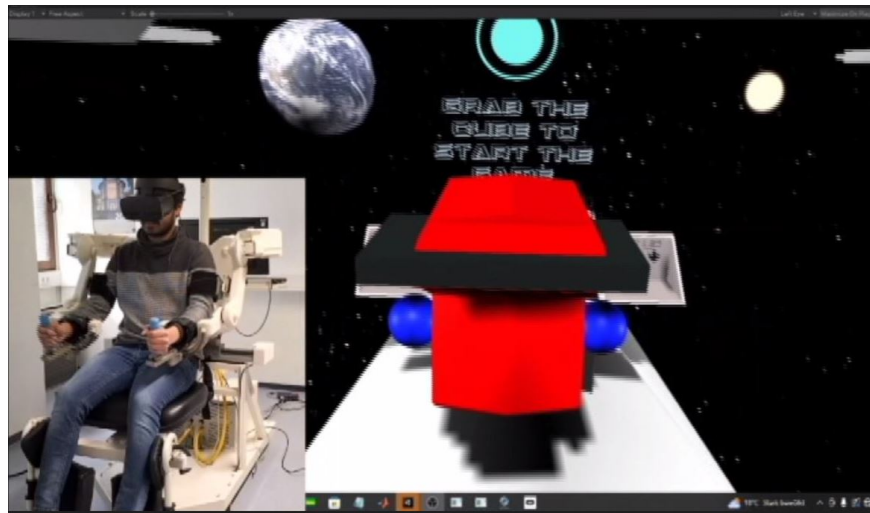


Figure 2.10 Unity Software

2. **MATLAB:** MATLAB 2.11 is software that is used for programming, design, and various types of engineering aspects in order to execute particular tasks according to its given rules and ideas and apply algorithms. It is mostly used for Mathematical computations and simulations. In our case, it is used for data analysis and to conclude which modality is better compared to the other. The Matlab version I used is R2023a and is useful for academic purposes.

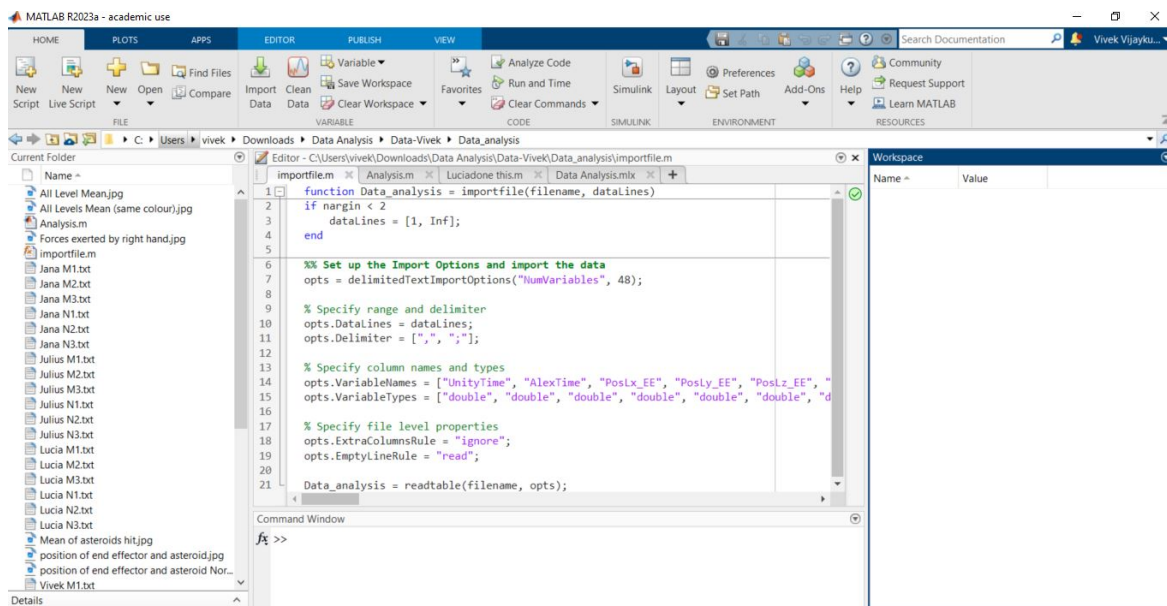


Figure 2.11 Matlab Workspace

Part II

Section Two

Chapter 3

Material and methods

3.1 Methodology

I would like to test the mirror modality to see if this would be adopted in a person with a stroke as we collaborate with the Tübingen University Hospital. To do this I want to do an experiment on 6 healthy subjects with no neurological problems and all right-handed as it is the dominant experimental protocol. I will ask the healthy subjects to run the Mirror Haptic Feedback Modality or the normal Haptic Feedback Modality as it is randomized so the subjects could start randomly with modality 1 or modality 2 by using only their dominant hand and each level will last for 2 minutes and later on I will compare this two modalities. Testing Modality 1 which is the Haptic Feedback and Modality 2 which is the Mirror Haptic Feedback will be helpful to see if the performance is comparable or if it is completely different to record in both modalities. I would like to see if the modality that I implemented is suitable for the user or if our modality is difficult to do.

3.2 Algorithm

The algorithm of the Exergame will go in such a way that There will be three levels with increasing velocity (speed) of the asteroids with targets of different dimensions.

Level 1:

In this level, the asteroids will spawn slowly with a spawning rate of 7 per second and a speed of 5 m/s on the X-axis from the right to the left. You will have to shoot them with your gun in the given time which is 2 minutes that will be displayed on a timer in the game.

The patient will have enough time to aim and shoot accurately as there will be few asteroids spawning in the environment.

Level 2:

In this level, the asteroids will spawn at a medium speed with a spawning rate of 5 per second and a speed of 10 m/s on the X-axis from the right to the left. You will have to shoot them with your gun in the given time which is 2 minutes that will be displayed on a timer in the game. The patient will have to be more accurate with their shots as the asteroids will move more quickly and be more challenging to hit as there will be more asteroids compared to the last level in the environment.

Level 3:

In this level, the asteroids will spawn at a high speed with a spawning rate of 3 per second and a speed of 15 m/s on the X-axis from the right to the left. You will have to shoot them with your gun in the given time which is 2 minutes that will be displayed on a timer in the game. This is the most challenging level, where the asteroids will move much faster with smaller dimensions. So, the patient must be very accurate with their shots as there will be many more asteroids compared to the first two levels in the environment which will be difficult for the target to shoot.

3.3 Materials and Equipment

Participants will experiment with the Exoskeleton (Alex) with 6 DOF, Oculus RIFT S which is a Virtual Reality Headset, and the space shooting game which will be a custom-made game designed for this study in the Unity Software using C# Programming language in Visual Studio.

3.4 Experimental Setup

The participants were 6 healthy unimpaired subjects 3 male and 3 female 3.1. They participated in this experiment as in the list below, They will be told the guidelines and the two measures will be taken.

Table 3.1 Demographics

Subject	age	Gender	Dominant Hand
Sub 1	25	F	Right
Sub 2	24	F	Right
Sub 3	24	M	Right
Sub 4	29	M	Right
Sub 5	28	F	Right
Sub 6	32	M	Right

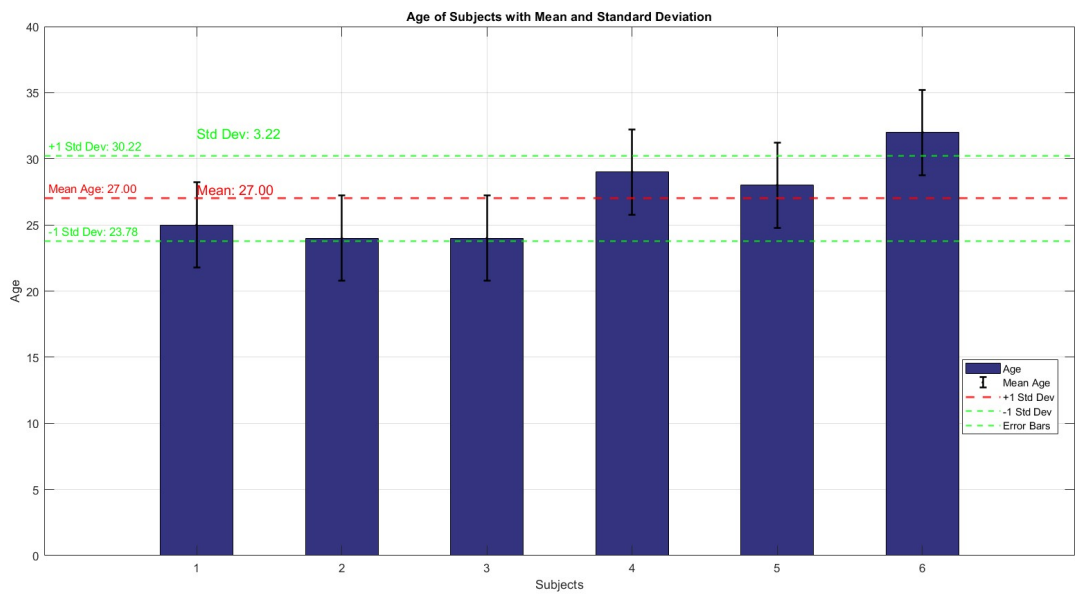


Figure 3.1 The average age was 27 years and the standard deviation was 3.22 years

Pre-Game Measures:

Before playing the Exergame, participants will complete a brief survey to collect Demographic information 3.1 and information about their video game experience. Then we will ask participants to perform tasks with the preferred hand (dominant hand) either right or left depending on their handedness.

Post-Game Measures:

After playing the game, participants will be given a 5-minute break and then complete a brief survey to rate their perceived difficulty, enjoyment, and frustration.

3.5 Experimental Protocol

Initially, the participant must try both modalities to get familiar with the game. This session will be done before the experiment starts as it will allow the participants to learn about the modalities and become comfortable using them, which can improve the accuracy and reliability of the results obtained from the experiment. The participant will use the exoskeleton to control the two end-effectors in the Virtual environment representing the participant's hands. Then when the game starts there will be a gun in front of the participant which they have to grab. After the gun is grabbed, they must aim toward the asteroid and move both hands forward to shoot and both hands backward to stop shooting. In Haptic Feedback Modality, the gameplay will be in such a way that both hands must apply enough force to keep the gun grabbed and if they are unable to do so the gun will either slip and fall or if too much force is applied to keep hold of the gun it will break into pieces.

In Mirror Haptic Feedback Modality, the gameplay will be in such a way that when both hands are in the position to grab the gun a link will be established between the two hands. The participant will then use the dominant hand to shoot the moving asteroids in the environment and the other hand will follow. You will get points every time you shoot the asteroid in the given time (2 minutes per level). The difficulty of the game will increase with each level.

NOTE – ALL THE LEVELS AND ITS DIFFICULTIES WILL BE DONE IN A SAFE SPACE AND VELOCITY WITH RESPECT TO THE EXOSKELETON WHICH IS VERY DELICATE AND SHOULD BE USED UNDER THE SUPERVISOR

Data Collection:

The following data will be collected during the Exergame: - Number of asteroids destroyed (Score). - Accuracy of shots (Evaluating the accuracy of shooting the target). - comparing the Motions of the exoskeleton in each level (Kinematics) with the position of Asteroids that are destroyed. - Force applied to aim and shoot the asteroids in each Levels (Forces).

Performance Measures:

Performance measures will be compared between the three levels using the data. And the comparison of normal Modality and the Mirror modality will be done.

Physiological Measures:

Physical symptoms if occurred during or after the game will be compared.

Ethical Considerations:

Participants will be provided with a questionnaire form and debriefed after the study regarding the aftereffects of the game. They will be informed that they can withdraw from the experiment at any time if they feel any discomfort.

Timeline:

1. Pre-Game Measures: 10 minutes
2. Modality 1 Game Play: 6 minutes
3. Modality 2 Game Play: 6 minutes
4. Post-Game Measures: 5 minutes
5. Data Analysis: 2-3 weeks

In this way, The study will provide evidence of the effectiveness of the Space Shooting Exergame with Robotic Mirror therapy. I would like to see if the mirror modality is difficult for the user so that the performances are better compared to the normal Modality and finally the results can be used to develop new therapies and interventions to improve the motor learning of the user.

3.6 Development and Integration of Exergame

To start with the development and the integration of the Exergame into the system was the interesting part to work on as the expectations for the concepts were like To apply the mirror modality to grab an object (gun) the hand is locked but not mirrored so you only use the mirror modality to grab. Then, you connect to the system (two hands pushing together) using the idea of the proportional derivative link spring and damper. We use mirror modality only with the intention of grabbing the gun once you grab the gun and want to move in the same direction, the arm is locked but not mirrored. Later on when Moving the object is not in the mirror modality it is locked with the other hand

Once you lock one hand on the gun and the other hand is locked and connected, it should switch to the dominant hand depending on the option. When locking the gun and switching the control from mirror to switch, a constraint will occur before we can move. So here what we have done is design a system. The system Alex is which you can buy ?? It's from an Italian company Wearable Robotics that produces this kind of bimanual manipulator. what we have done is, we have created haptics inside which is the first example of bimanual manipulation.

So what does it mean? It means that when you grab the object the force that is calculated inside the object by the graphical engine is then sent inside the controller at the low level of the system that shares the force between the two hands. It's called bimanual coupled manipulation and this is something like haptics which was not done before but in the Aries Lab. You can manipulate or feel the impedance of the resistance in the virtual environment with one hand. But when you have two hands they converge on the same mechanical system. You have the calculated Dynamics that is produced when you manipulate the object and rendering in terms of force fields. What we are doing is just using this kind of approach for Mirror Therapy. For example, if one hand is impaired and the other is a paretic hand when I start moving my hand they are all mirrored and when we converge towards the object as soon as I touch the object the object is locked with the two hands and then your impaired hand moves. When the impaired hand moves it moves with a certain amount of force which is 5N for safe space and if the force exceeds the object breaks and the game will restart. The force should move in One Direction despite you have the assistant in your hand. So let's suppose the one is locked and the other one is moving hand. Here we can move the hand but cannot push it as it follows the impaired hand because it is locked. This is sort of a neural Theory. This is also tested with the Tübingen University Hospital. So the graphical user interface low-level control haptic rendering is the output.

3.6.1 C# Alex Script

the Alex script was in function for the system to work. so we began from this script as this was the main function to control and work with the system so we started understanding the script and further developing the Exergame to fit the criteria of the project. so the description of the script is as follows:

using System.Collections.Generic; This command contains interfaces and classes that

define the collections.

public class Alex : MonoBehaviour this unity script derives and initialize the data types.

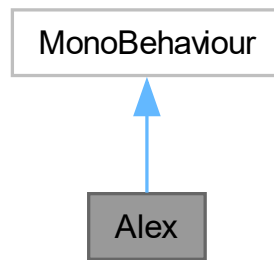


Figure 3.2 class reference

Void start is called before the first frame update. It is a function where you assign the game objects to the variables. To be sure the Script works fine we need to have all the game objects. The Shared memory is needed to communicate with Alex. In Alex script the left hand, right hand, virtual ones, Mano-destra and sinistra are the spheres. Visual L and R are another spheres which are part of the end effector. For the table if we grab the gun and if the forces are bigger than 15 newtons they just deactivate and the table disappears. Mano-sinistra and destra are more important parts in the scripts because we get the position of the end effector and we translate it to the position in the game. Most of the objects you need in the game are in the Alex game objects. For fixing the hands it's the same just a flag attribute (enum) where you fix it it's in the CASE that if you grab the gun and then set the hands to the position of the cube.

Update is used for regular updates such as moving physics objects in the game

Fixed update is used for regular updates such as adjusting physics rigid body objects.

SHM is the shared memory.

Time delta time; is the completion time in seconds since the last frame .

Quaternion: is Used to represent a rotations.

Void set saturation limit is to set the safe space for the subject to perform the experiment.

To keep the code clean I had to remove the unwanted parts and comment on the parts that

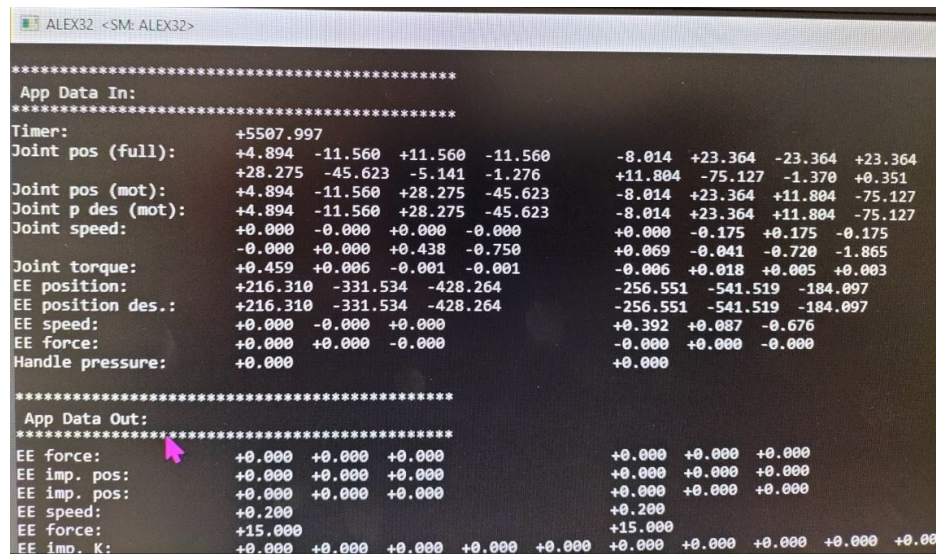


Figure 3.3 Real Time Data of The Exoskeleton

were important for the references.

Then to understand the information parameters in Alex 32 we see the real-time data when the moment of the end effector is done and is displayed in the shared memory and thus we can write the value of the position in the console. For example, When you compute the force during interaction we can use parameters such as stiffness, EE, and impedance. the following two figures show the data in DataInStruct and the DataoutStruct that is The structure contains all the information related to both the arms of the exoskeleton and allows the setting of all the parameters related to both arms of the exoskeletons.

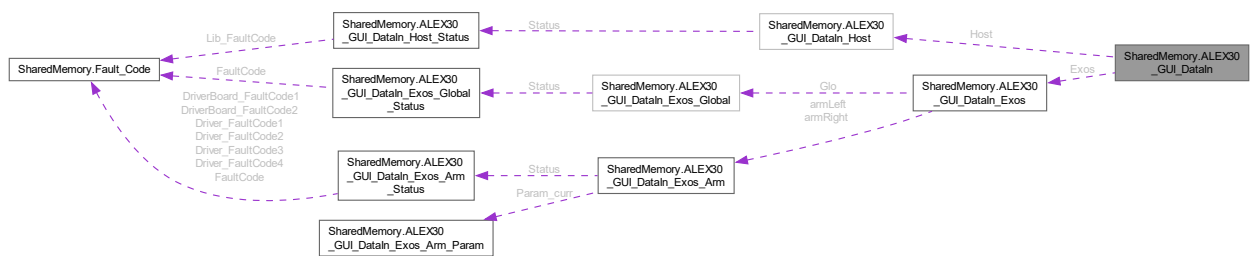


Figure 3.4 App Data In is the data that we read in real time

so in order to start with the mirror modality first we need to fix the object that is the gun in

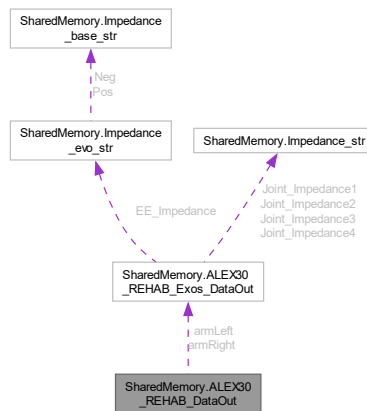


Figure 3.5 App Data Out is the data that we write in the console

my case. where you control only one hand either Right or Left whichever is dominant then read the force from one hand to the other via shared memory.

3.6.2 Limitations and challenges

In my study, we are not only using the old modality which is the Haptic feedback modality but the idea is to create a mirror modality with haptic feedback as the mirror modality of the system which already exists works only in the plane environment and is limited as it had some limited restrictions which made it difficult to perform in the Exergame. It means that the movement was limited and it created symmetric patterns. It resulted in an error in the exergame where we have to perform the Bimanual task using more degrees of freedom. The mirror pattern in the system was in such a way that when I tried to perform a movement to the right the lateral divergence occurred which means if I moved my dominant hand in the right direction the non-dominant hand moved opposite of it that is in the left direction. and if I tried to perform a movement to the left, lateral convergence occurred which means if I moved my dominant hand in the left direction the the non-dominant hand moved closer reducing the gap between both hands and making the Exercise difficult as the purpose is to grab the gun and move it around in the virtual space. so we need to implement other directions in the environment or just the central as it is in order to fix it. Due to these limitations, it became challenging to implement the robotic mirror therapy in the system but as a Robotics engineer, this kind of challenge can be overcome from a kinematics point of view taking into account

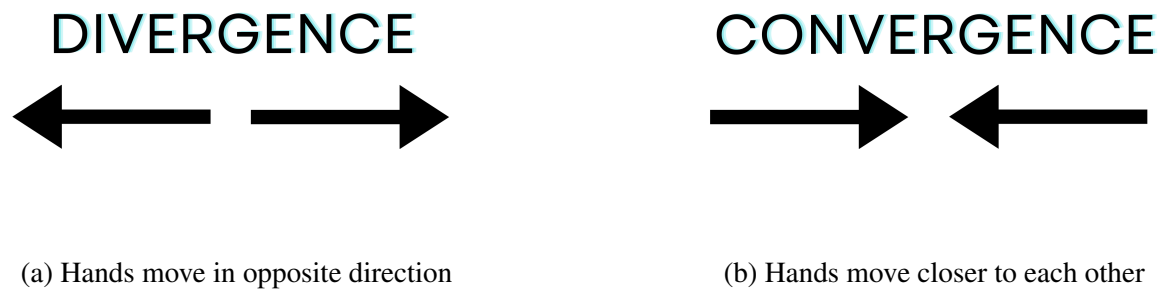


Figure 3.6 Limitations in the Mirror modality of the system (the above images are created in (<https://www.canva.com/>))

what is to be obtained from the results.



1 Lateral Divergence



2 Lateral Convergence

Figure 3.7 Movement in X-axis

As you can see in this figure 1 Lateral divergence occurs when you try to move your dominant hand to its side or away from the non-dominant hand. which means if you have to move the gun bimanually in the virtual space to the right or left depending on your dominant hand it will not be possible due to these limitations as seen in the above figure 3.6a. And in the figure 2 lateral convergence occurs as you try to move your dominant hand to the other side or towards the non-dominant hand. which means if you have to grab the gun bimanually in the virtual space it will be possible as the distance between two hands decrease as seen in this figure 3.6b but you cannot move the grabbed gun in the virtual space to shoot the asteroids which require the freedom to move around and assist the upper limbs for rehabilitation in the safe space.



Upward Movement in Y-axis



Downward Movement in X-axis

Figure 3.8 Motions in Y axis

As seen in the above two figures the vertical movements in the system works well as expected but again the problem is for moving around in the Virtual environment in order to shoot the asteroids and this again is the constraint which stops us to fulfil the goal of this Exergame so what we did

3.6.3 Fixing the object (gun)

The idea about fixing the gun is to set a point for the hands to get fixed and linked to each other so they can be mirrored. Lets take an example, You decide on subject 1 whose right hand is the leading hand, so I try to move the gun which I have to move to shoot the asteroids, once I am on the object, I fix the other position. That means when the subject touches the gun by the right hand you fix the other hand on the object and move in the space, and you feel the force. There could be one Possibility for changing the environment as in the previous modality the task was just to pick and drop the object and through the object you received haptic feedback. Here the goal is to link both hands after it reaches the cube so once you start the experiment we tell the subject to only use one hand which could possibly be their dominant hand and to grab the object which is the gun in the VR scene and after grabbing the gun both hands are locked to be in the mirror modality. The two blue spheres in 3.9 represents the position of the end effector. In the beginning, the first task is to fix the object that is the gun (in black colour 3.9) then we can read the data of the position and force of the end effector of the dominant hand and write it on the other hand which will follow. So we do it from the code to set the position which is Fixing Permission

Once the dominant hand and object is fixed you have to write from the code that only the dominant hand can guide in terms of force. In terms of the movement, we need to understand



Figure 3.9 Gun to grab and start the Exergame. The two blue spheres represent the position of the end effector.

how to create the link between two hands. So if you are in the wearable mode which is one of the functions of Alex-rs in this mode the exercise can be done without anyone being inside the exoskeleton for working purposes you move only the right hand and you write on the left hand the control of the right hand which we can see from the shared memory. The problem was the link. So, for the link, we can provide the same position and we can do it by fixing the right hand and the right forces of the end effector and then the position of the right hand on the left so in this way the left hand will follow the right hand. so you compete the forces. as usual but you overlook the force in the left hand and instead, you only read the position of the left end effector. So, in this way in place of showing the position of the left hand, we show (write) the position of the right hand. In simple words, we get the solution to link the hands if we write the position of the right hand on the left end effector in order to make the left hand follow the right hand.

To fix the cube in the c# script I added a child class to The gun where the child has a collider 3.10 so if the sphere touches or is triggered at that particular point it will collide with

it.

The basic idea was to grab the object with the dominant hand and how I did is I wrote a script where I got a reference to the Alex script and then I assigned it. If something enters the hitbox area in the center of the gun and if it is a dominant hand so I set the fixing permission to true or else it's false as in this command **void setFixingPermission** which is the bool function (bool permission). Then in the In Alex script I added cube fix **getCubeFixed ()**

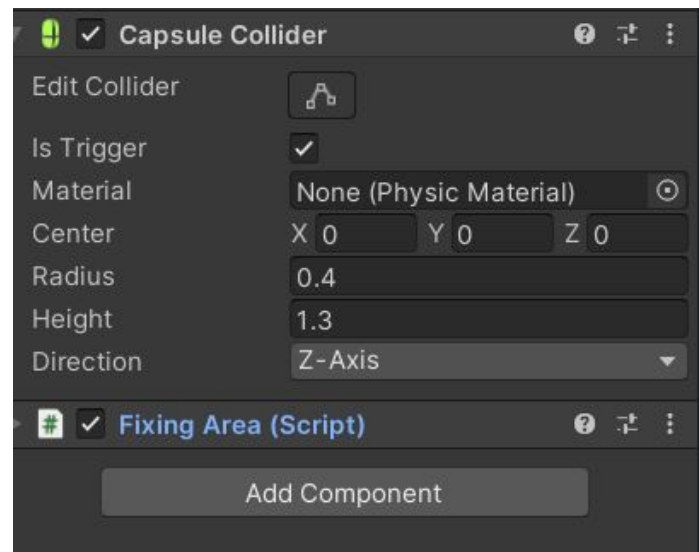


Figure 3.10 Parameters of the child class which has a collider.

here If the force from the right hand to the left hand is big enough then pressing the gun in it will snap in and I will be allowed to fix which I have set to the offset. Offset is the distance between the Hand and the center of the gun. If the gun is fixed I have set the position of the gun in the center to the position of the Hand in addition to the offset and give the offset of the right hand and multiplied by 2 So in this way If the middle of the cube is 1 unit apart from the right hand then the left hand will be 2 units apart to make it symmetric.

3.6.4 Mirror Modality Implementation

To implement the Mirror modality where your dominant hand leads and the non-dominant hand follows we first give the [SerializeField] attribute in the Alex script so it is visible in the unity editor and changed according to the preference of the subject 3.11 which is a bool function in the script. the code is written in the following way where there is an **IF Condition** and according to the selected hand by the subject the game goes on and that particular hand will lead in the context of shooting the asteroid.

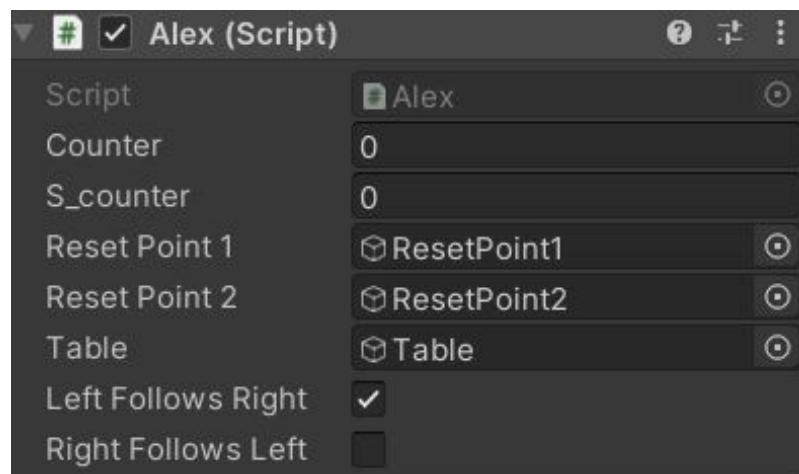


Figure 3.11 his is where the subject chooses which hand to lead.

```

    if (cube_fixed)
    {
        if(leftFollowsRight && rightFollowsLeft)
        {
            box.transform.position = manodestra.transform.position + CubeOffset;
            vl.transform.position = manodestra.transform.position + new Vector3;
            vr.transform.position = manosinistra.transform.position - new Vector3;
            visual_R.transform.position = manodestra.transform.position;
        }
        else if(rightFollowsLeft)
        {
            box.transform.position = manosinistra.transform.position + CubeOffset;
            vr.transform.position = manosinistra.transform.position + new Vector3;
            vl.transform.position = manosinistra.transform.position;
        }
        else if(leftFollowsRight)
        {
            box.transform.position = manodestra.transform.position + CubeOffset;
            vl.transform.position = manodestra.transform.position + new Vector3;
            vr.transform.position = manodestra.transform.position;
        }
    }
}

```

the above code works in such a way that it will update the left and right hands depending on the calculations with the data from the shared memory depending on which logic is true from these conditions which are `rightFollowsLeft` and `leftFollowsRight`. if the right hand follows the left then the `manosinistra` which is the left-hand sphere representing the left-end effector's position of the exoskeleton data will be read from the shared memory and will be written on the right hand and if the left hand follows the right then the `manodestra` which is the right-hand sphere representing the right end effectors position of the exoskeleton data will be read from the shared memory and will be written on the left hand. It means that if the cube is fixed and according to the bool function if it is true the respective hand will become a leading hand and the other will follow

The other functions are **Void start()** which is a unity function which is called one time at the beginning of the game

Void update() update function is called every frame.

Void Late update() function is also called every frame but after the update function.

Void Fixed update() has a fixed time set, in Physics functionality it is used for collision or speed change.

Void Fixed update() creates a text in the file for collecting the data every 0.01 Seconds after 1 second.

void update() time.real time since startup >7) wait 7 seconds to set active the plane front so we deactivate it in seven seconds.

void fixed update() adds the data from Alex and then we have a switch case. it's like an IF function but it has more possibilities IF function is like if you want to do this it's true ELSE do that. In the switch function depending on the value of the variable, we do different things such as using cases.

case 0= After 7 seconds the game starts and then we can grab the cube.

case 1= the cube is fixed and the mirror modality is active.

case 2= if the game ends or if the cube has fallen in the Normal modality the reset point is active and the game starts again.

For the Object fixed part, we take the position of the box and set the position of the right hand and also the offset. Same for VS left hand we take the position of the right hand and get offset two times, one time the offset is in the middle of the gun and the other time the offset is in the outer side of The gun.

Gun Firing For firing the gun and shooting the asteroids i implemented a method called **adaptLaserGun()** as there is no proper button or place to fire the gun normally. This was

my idea to control the firing of the gun and this could be easier to do this task Bimanually so when you move your hand forward in the z-axis it will fire and if you take it behind it will stop firing .Basically, it controls the way how the subject fires the gun in order to shoot the asteroid. It is based on the position of the end effector that if it is greater than 0.3 in Z-Axis it will trigger the firing of the gun in case 1 only when the hands are fixed on the gun. The code snippet is as below.

```
private void adaptLaserGun()
{
    if (manosinistra.transform.position.z > 0.3)
    {
        MuzzleFlash.Play();
        gun.LaserGunFired();
    }
}
```

In conclusion of the game development, I implemented the game. The game is related to the device. Basically, I checked every condition that was fine to allow the correct movements. I implement the asteroid shooting game with the virtual device. I scaled the virtual reality environment to the device in a way to creates a proportion between the user and the environment. In the mirror modality, I worked in the lab with a colleague to implement this mirror modality and the results were positive in the performances of the subjects who undertook the experiments. Thus this kind of therapy proves to be a solution for cost-effective medical facilities and self-paced rehabilitation techniques (19).

Chapter 4

Data Analysis

4.1 Background and Scope of the Data

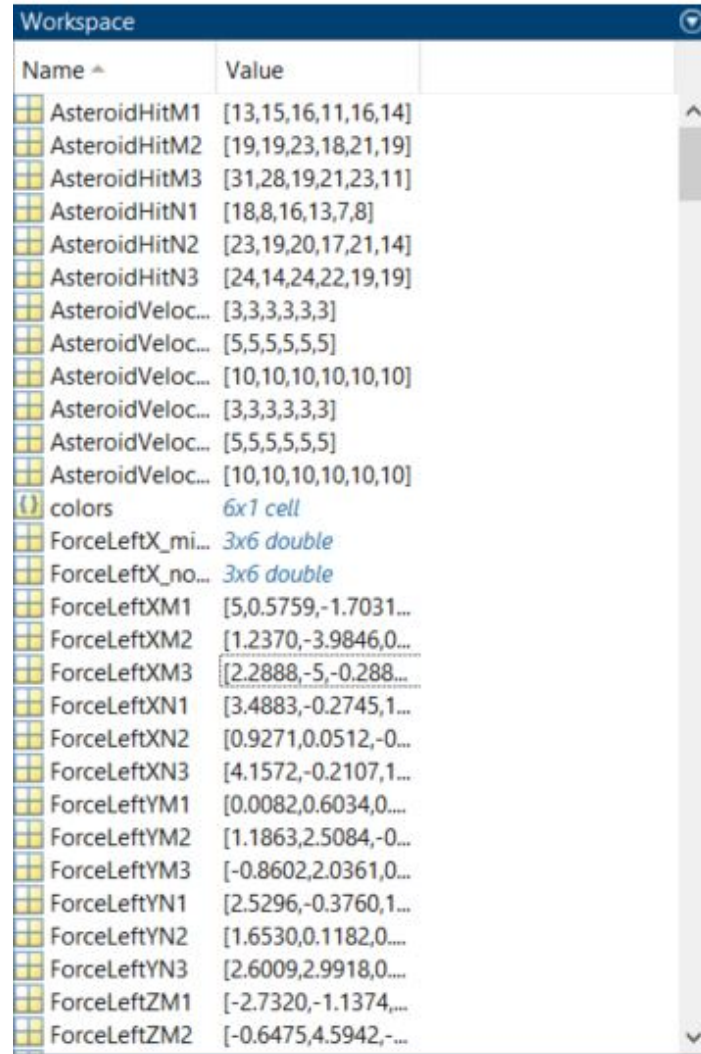
Visualization is an important part of the data analysis and if the data is right and is organized in a right way the predictions are nearly accurate (20). For the data analysis part, the first step is to collect the data of the participants who did the experiments. I used the import file function where you can write the file name and we get the output of the matrix or the struct data that we see in the workspace tab 4.1. I took the data from the beginning to the end which is approximately 2 minutes per level then the next step is to import the data filter it and then compare it between subjects in the same level Mean+-Standard deviation for all the vectors and columns If we want to check the velocity of the end effector in Y-axis then you get the specific column containing the data and then select the vectors of all your subject and put it in one matrix. the vectors you are collecting must be filtered and of the same size otherwise, they can't be in the same matrix. Then you can make the mean for example you get one mean velocity vector for all the subjects in that particular level and the standard deviation as the comparison should be between subjects for the same level. This is the function of MATLAB as I have used this Software for analyzing the data and concluding the results.

To collect the data we write a small code in the Alex script to create a text file where all the data is collected. In the code, we use the function called **StreamWriter** which is used for documentation which means we access something outside the program. It is a writer who writes a file (text). It's done with the function create text(this function is called every 0.1 seconds) and just creates some file or a path that says if there is no file it creates a file and if it already exists it writes down all the information of the position of left and right Arms are also the position of EE Alex from the SHM the Numbers are mostly vectors or arrays.

As below we can see how we write it in the script to obtain a text file where we get the data which we can further analyse.

```
void CreateText()
{
    if (!File.Exists(path))
    {
        File.WriteAllText(path, "Unity time; Alex Time; " +
            "PosLx_EE; PosLy_EE; PosLz_EE; PosLx_virtual; PosLy_virtual; PosLz_virtual;" +
            "PosRx_EE; PosRy_EE; PosRz_EE; PosRx_virtual; PosRy_virtual; PosRz_virtual;" +
            "VelLx_EE; VelLy_EE; VelLz_EE; " +
            "VelRx_EE; VelRy_EE; VelRz_EE; " +
            "AA_Shoulder_Left; PS_Shoulder_Left; FE_Shoulder_Left; FE_Elbow_Left; " +
            "AA_Shoulder_Right; PS_Shoulder_Right; FE_Shoulder_Right;
            FE_Elbow_Right; " +
            "Fx_L; Fy_L; Fz_L; " +
            "Fx_R; Fy_R; Fz_R; " +
            "playerScore; asteroidshit; " +
            "Asteroid Velocity; PositionAstX; PositionAstY; PositionAstZ \n");
    }
}
```

After the experiments which were done on 6 subjects I have to make sure that the frequency and amplitude frequency of the collected data were correct in order to analyze it better. I did two sets of dual tasks for helping the subjects in the 3-day trial to get used to the game in both modalities. As there is not a lot to compare I define the parameters of the subject for example the scores the forces and the position of the end effectors for each level. I saved the scores for each subject and checked the accuracy of the moment. then the Level of the asteroid and the level of the end effector to check if you shot the asteroid correctly to have the reference of the height of the asteroid also with the measurements of the Kinematic performance to see if the end effector went correctly towards the target or not as the asteroids come from different positions and I can detect the height of the asteroid which is the target to shoot so that I can know the movement of that subject is accurate and appropriate for the motion as the gun is fixed and the subject is supposed to move around.



Name ^	Value
AsteroidHitM1	[13,15,16,11,16,14]
AsteroidHitM2	[19,19,23,18,21,19]
AsteroidHitM3	[31,28,19,21,23,11]
AsteroidHitN1	[18,8,16,13,7,8]
AsteroidHitN2	[23,19,20,17,21,14]
AsteroidHitN3	[24,14,24,22,19,19]
AsteroidVeloc...	[3,3,3,3,3,3]
AsteroidVeloc...	[5,5,5,5,5,5]
AsteroidVeloc...	[10,10,10,10,10,10]
AsteroidVeloc...	[3,3,3,3,3,3]
AsteroidVeloc...	[5,5,5,5,5,5]
AsteroidVeloc...	[10,10,10,10,10,10]
colors	6x1 cell
ForceLeftX_mi...	3x6 double
ForceLeftX_no...	3x6 double
ForceLeftXM1	[5,0.5759,-1.7031...
ForceLeftXM2	[1.2370,-3.9846,0...
ForceLeftXM3	[2.2888,-5,-0.288...
ForceLeftXN1	[3.4883,-0.2745,1...
ForceLeftXN2	[0.9271,0.0512,-0...
ForceLeftXN3	[4.1572,-0.2107,1...
ForceLeftYM1	[0.0082,0.6034,0...
ForceLeftYM2	[1.1863,2.5084,-0...
ForceLeftYM3	[-0.8602,2.0361,0...
ForceLeftYN1	[2.5296,-0.3760,1...
ForceLeftYN2	[1.6530,0.1182,0...
ForceLeftYN3	[2.6009,2.9918,0...
ForceLeftZM1	[-2.7320,-1.1374,...
ForceLeftZM2	[-0.6475,4.5942,-...

Figure 4.1 Data in the MATLAB Workspace.

First I check the data is in which folder. then I make a string vector and then I check the size and not the length in the vectors after this I check for the files of name and type so I can put the name of the index (subject Vivek) and then use the function for calling filename and then I remove all the null data with `r missing` function **Data=rmmissing(Data);** which removes the rows and columns which are missing. For all THE mirror 1 2 3 and Normal 1 2 3 levels we take a big matrix with each column with the data from each subject that is 6 subjects with multiple rows and 6 columns. for example, from the 6 velocity vectors that is one for each subject, I make the mean and I get one velocity for all the experiments in mirror level 1 and so on I can reuse the code for other levels and if I want to get all the subjects I can

do it individually. There is also a function called **Is Change** this function detects the change in the data. In My case, I have to check when my first asteroid was shot. so you put in the parenthesis the vector or the matrix to check the moment in which you change from 0 to 1.

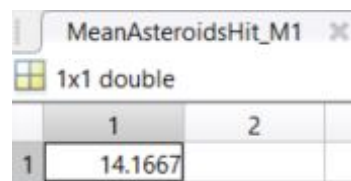
4.2 Importing the data in MATLAB

To import the data I used this import function in MATLAB.

function Data_analysis = importfile(filename, dataLines). This function imports the required data and further, I can reuse this function to import the other related data for the analysis as shown in the below code.

```
JanaM1=importfile('Jana M1.txt');
JanaM1=rmmissing(JanaM1);
JanaMirror1 = JanaM1(:, [1,3:5,9:11,37:48]);
p2_mirror_level1 = JanaMirror1(901:end, :);
```

I did this for all 6 subjects and with all three levels of Mirror 1, 2, and 3 and Normal 1,2, and 3. it's best to add all the data in one folder with a different name to avoid any confusion. Then I have to make a big matrix with each column with the data from one subject. so, if I have 6 subjects then my matrix is going to be with multiple rows and 6 columns which would be one for each subject and then you make a mean of these values. For example, out of 6 AsteroidsHit_M1 vectors 4.1 that is one for each subject, I make the mean so I have got one MeanAsteroidsHit_M1 for all the experiments in mirror level 1. In the import file function change the double and the header file name for each vector to get all the data. Once



MeanAsteroidsHit_M1	
1x1 double	
	14.1667

Figure 4.2 MeanAsteroidsHit_M1

you import you get the data in the workspace and you can open it and when you click on it you get the matrix without the labels for example if you want to check the Y position of the asteroid then the column you need is the 18 in the table and when you have to plot that then you can type plot (**Data.data(:,18)**) as it is a struct type and the dot is functioning between the file name and the specific file name we are looking for.

Table 4.1 Position Vectors

16	17	18	19
AsteroidVelocity	PositionAstX	PositionAstY	PositionAstZ
3	0.8440	-1.8223	14.8411

4.3 Parameters to consider

The data was taken from the beginning of the game to the end which is approximately 2 minutes for each level. Forces and trajectories were recorded at 200 Hz and offline filtered using a 6th-order low-pass Butterworth filter with a 10 Hz cutoff frequency. The data which is used for the data analysis is as follows.

1. **The asteroid shots** This is the number that gives us an idea of how many shots were successful in each modality as it gives us the approximation of which participants perform better.
2. **Height of the asteroid and the position of the end effectors.** This is the data to compare the kinematics performance with the virtual environment.
3. **Position of the first asteroid and the others** This data will give us an idea of the starting point in each modality and which of it impacts the shooting position.
4. **Overall To check the Forces, Scores, Asteroid Hits, Mean, and the Standard deviation.** Overall we check this data and we plot different figures to understand it visually.

4.4 Results

Overall the subjects performed well in the Mirror modality when compared to the Normal modality because of the mean and the standard deviation of each parameter that was recorded. As there were differences in both the modalities 4.3 we got a mean of all the scores and we can clearly see the mirror modality has a higher number of scores. In addition, I also want to see the different parameters as discussed above to check the comparison between both modalities. the ones that I want to check are the position of the end effectors and the position of the asteroid in both modalities of level 3 as it is the last and the toughest level to play and we can therefore determine in which modality the subjects performed better.

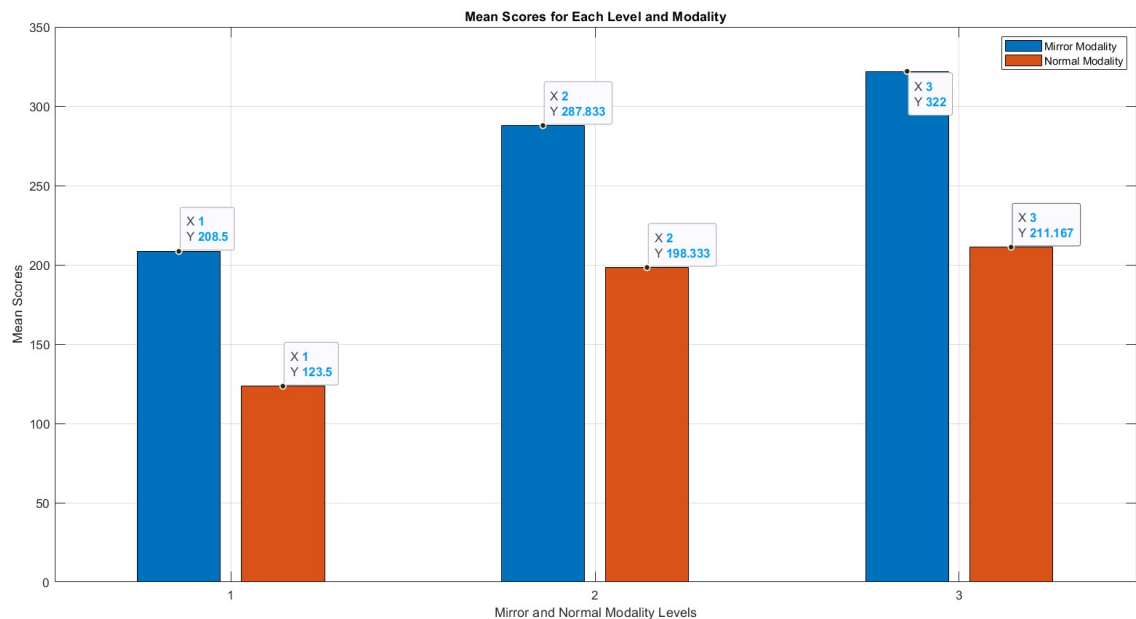


Figure 4.3 The figure represents the Mean scores of all subjects in all levels in each modality of the game. The scores of all three levels in the mirror modality are higher than 215 where the mean scores in mirror modality level 1 is 208 in level 2 is 288 and level 3 is 322 which is the highest and the mean scores in the normal modality in level 1 is 123 which is the lowest and level 2 198 and in level 3 211

Statistical Analysis

In this analysis, I can see after we get the mean and std deviation of the parameters such as scores and asteroid hits to make a comparison with the ANOVA test as the data was normally distributed we see if there are significant differences between the mirror and normal modality. Then we see if there is a statistically significant difference by looking for the value that is called a P-value, if it is less than 0.05 4.4 then there is a difference between the two groups of data that I am comparing. With the T-test we can compare the mean of the mirror and normal modality for each level so I used these tests with a p-value < 0.05 to check if there are significant differences in both modalities and the p-values were < 0.05 so we can say that the mirror modality was better compared to the normal modality. and finally we compare the two overall mean and standard deviation of the Mirror and normal modalities to confirm the significant difference 4.5

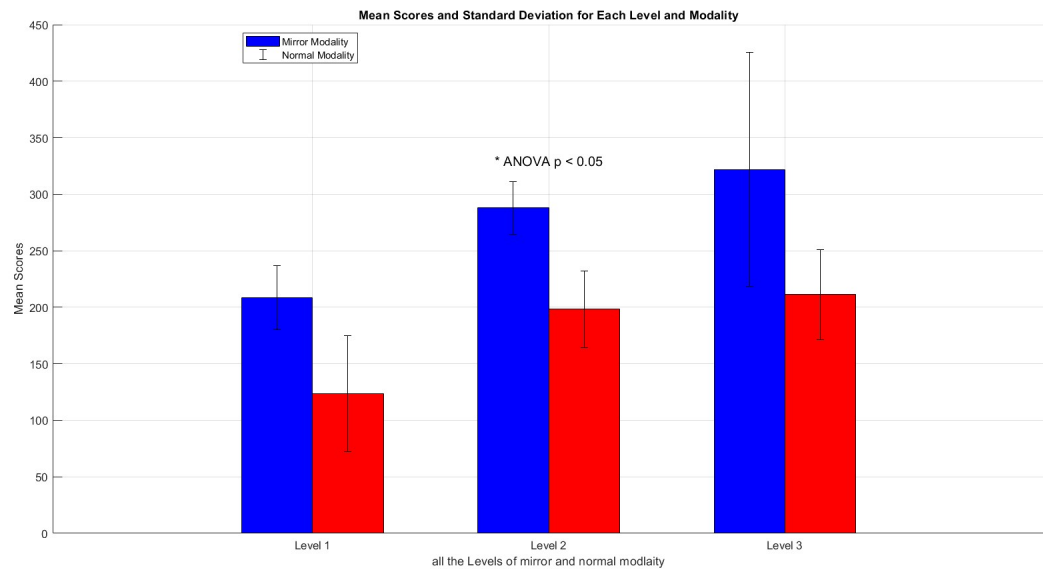


Figure 4.4 Here the plot represents the standard deviation of the scores and shows the variability in the data as we can see there is a significant difference using the ANOVA test

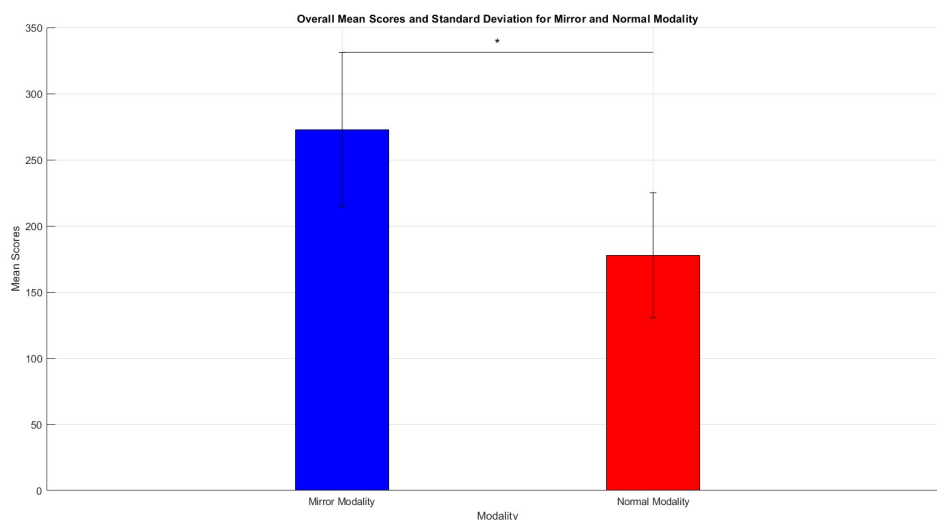


Figure 4.5 Here the plot represents significant difference between the mirror and the normal modality

Comparing the end effector position and the height of the asteroid

This is where we compare the kinematics performances and the virtual environment position of the end effector and the position of the asteroid and you can see in 4.6 the normal modality is visualized and in 4.7 Mirror modality is visualized. the target was not able to reach in

Normal modality which was a failure and we can determine whether the end effector and asteroid height match or else it does not match by referring the distance between the height of the target and end effector. As closer the distance, the better the performance.

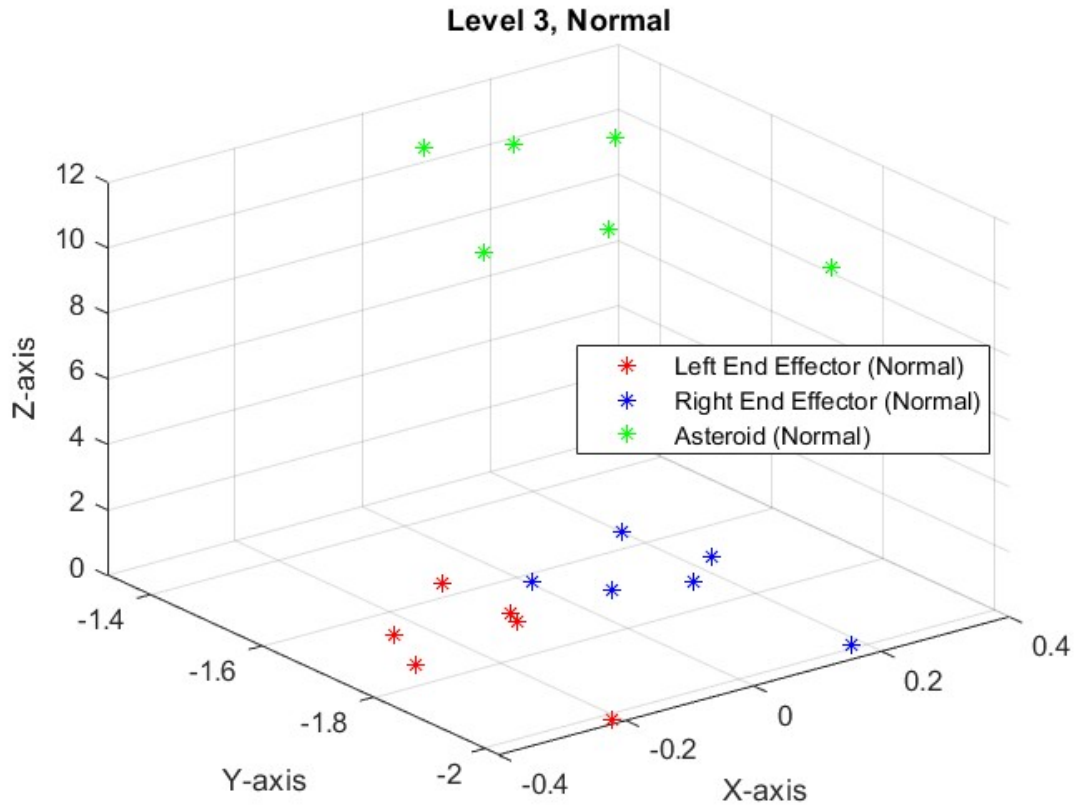


Figure 4.6 The figure represents the positions of the left and right end effectors (normal and mirrored) in 3D as well as the asteroid (normal) at a specific point in time ($t = 1.00192$) in normal modality.

The RGB star shows the trajectory of the left and right end effector and the asteroid spawning in the virtual environment and the space between them shows the movement of the asteroid and the end effector over time. This shows how the movement of the hand is made to shoot the target and how the movement affect the positions.

Initial Asteroid Positon

The Initial or the first asteroid is very important to understand the starting position of movement to shoot it. Here if you want to extract the information for example how you

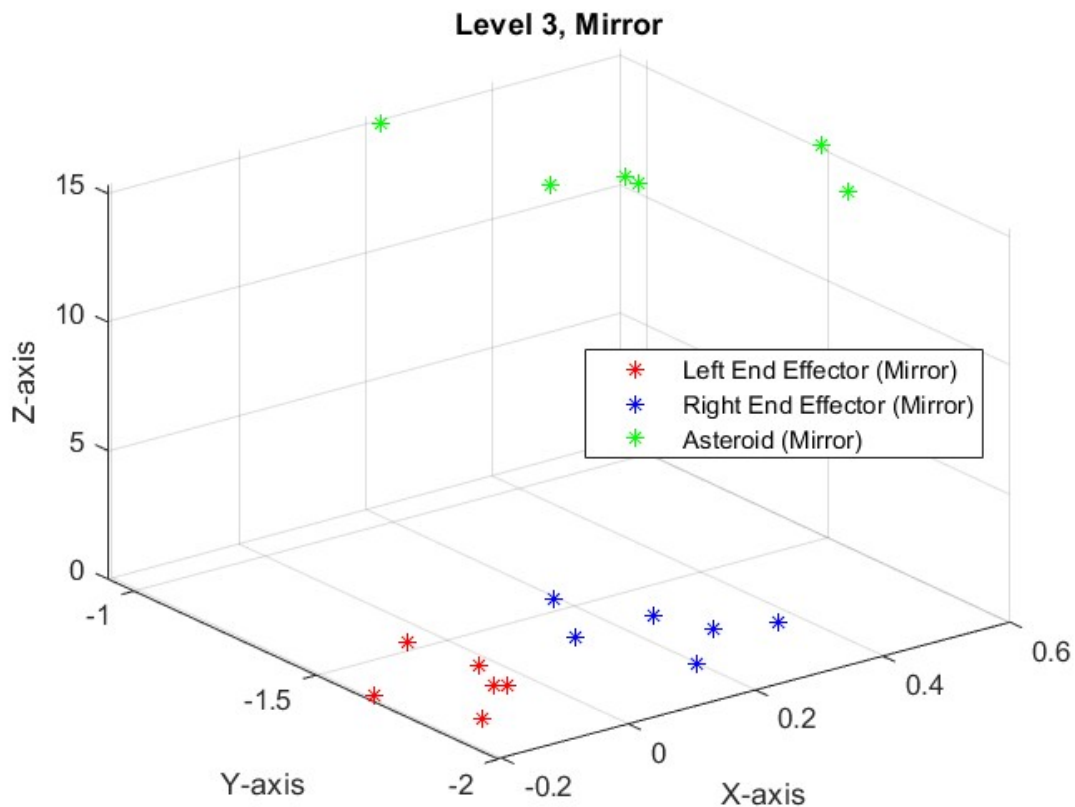


Figure 4.7 The figure represents the positions of the left and right end effectors (normal and mirrored) in 3D as well as the asteroid (normal) at a specific point in time ($t = 1.00192$) in Mirror modality.

move your hands to take the first asteroid with the other data you know in the plot 4.8 the position of the first asteroid and others as there are overall 10 asteroids marked as dots, where the blue dot represents the asteroid positions that were spawning in the virtual environment and the red dot, represents the first asteroid position. There are many ways to compare the performance but this is one way to analyze the overall flexibility and the performance of the subject. This may also give you an idea of where you can make the changes in the game and improve it for future development.

Asteroid hits and the error bar

When the rehabilitation game starts and the asteroid starts spawning the subject is supposed to hit those asteroids and thus from there the parameter `AsteroidsHit` is referred to as that hit. The figure 4.9 shows a visualization of the mean of how many asteroids were shot in both

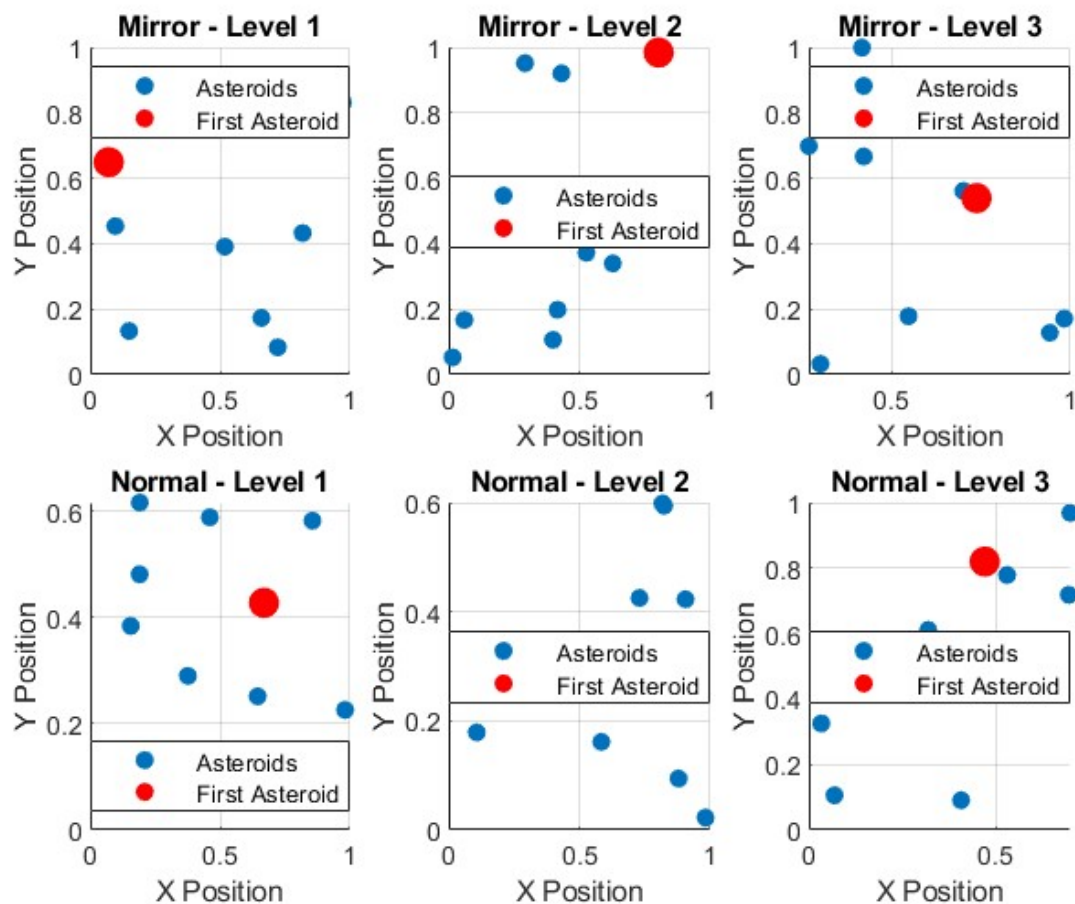


Figure 4.8 X-axis Represents the position of the asteroids moving in x coordinates, Y-axis Represents the position of the asteroids moving in y coordinates, Small filled circles represent the positions of multiple asteroids. and a Highlighted point is a larger red-filled circle that represents the first asteroid.

modalities and all levels and the variability in the data and also shows the standard deviation error bars. The unit measures are the number of asteroids shot in each level and each modality. The X-axis is labeled as Level, the Y-axis is labeled as Mean Asteroids Shot. The error bars are represented in two data series one is the mirror modality in blue bars and the other is the normal modality in red bars. the graph suggests that the number of asteroids shot may vary depending on the level and modality. It also shows that there might be some difference in the number of asteroids shot at each level between each modality, as indicated by the error bars. And as seen the Mirror modality is higher than the Normal modality. this shows that the subjects were having it easier to shoot more asteroids in the Mirror modality when we compare it to the normal modality in different conditions. Then the **MF** The maximum

failure was observed in the Normal Modality as seen in the figure 4.10 where the Red bar refers to the Normal modality and the Blue bar refers to the Mirror modality.

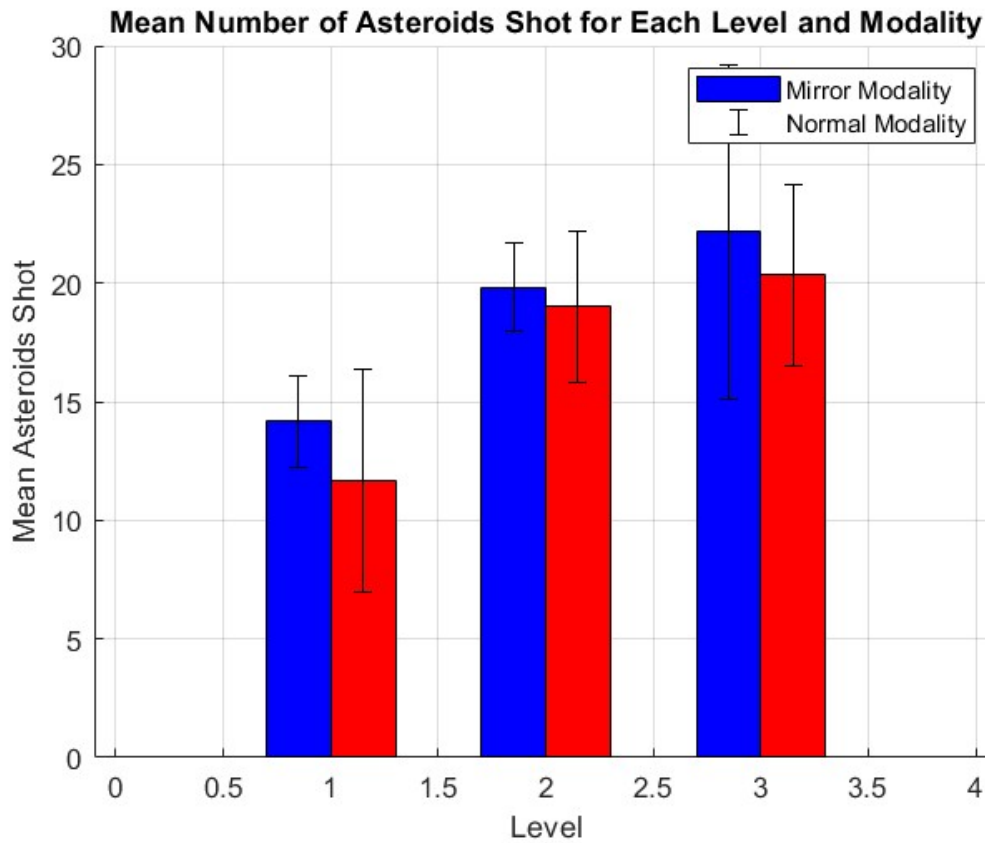


Figure 4.9 The error bars represent the standard deviation of the data for each level and modality. The standard deviation is a measure of how spread out the data points are around the mean. Smaller error bars indicate that performances were more consistent, while larger error bars indicate greater variability in the number of asteroids shot. The X-axis represents each Level and the Y-axis represents the asteroid shots that appear to range from 5 to 30, at intervals of 5.

Forces

Forces play a very vital role in this experiment as for every movement of the end effector there is enough force to be applied and thus we can distinguish which modality acquired more forces to finish the task. We can assume that if more force was applied in the normal modality to attempt and if less force was applied in the mirror modality to attempt the task that means it was hard for the user to perform the task in that particular modality. The dominant hand for each participant was the right hand and so we took the data for the right

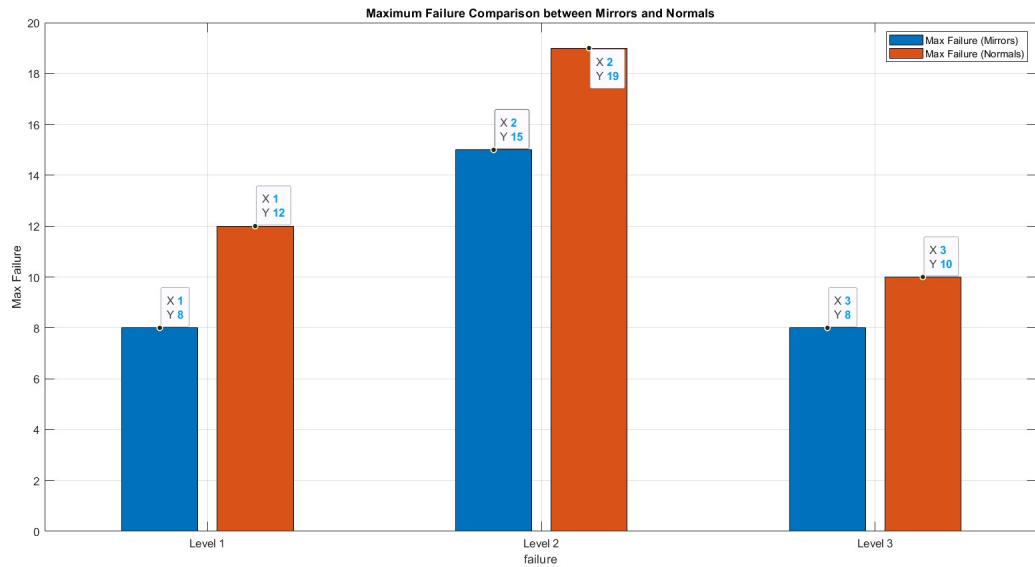


Figure 4.10 The X-axis represents the different levels and the Y-axis represents the Maximum failure to shoot the asteroid the failure in mirror modality level 1 is 8 level 2 is 15 level 3 is 8 whereas in normal modality level 1 is 12 level 2 is 19 a level 3 is 10.

hand in each axis of each level and the below figures 4.11 refer to the right hand in x-axis 4.12 refer to the right hand in Y-axis and 4.13 refer to the right hand in Z-axis and this all plots show the values and the scattered plot for the forces.

4.5 Discussion

This is an Implementation of bimanual coupled manipulation. In this project, we do VR and haptics. We find How to translate interaction in the VR to physical interaction by a robotic system. Translating the object that is the gun which has the weight that is simulated inside the graphic engine. Taking all the forces that are simulated into the engine and then transferring them into the exoskeleton. We feel the force to grasp the object (gun). The gun is to move, we do this thing for stroke rehabilitation because in this exercise one hand leads, and the other is controlled or mirrored by the other. The Dynamics of the object will be perceived in both hands. We promote the motion of the paretic limb. Once the object is grabbed, One arm is a dominant hand and the other is connected with the hand that is active that is your dominant hand and then you can move the object (Gun) by riding the non-dominant hand as well. If the non-dominant hand does not follow then you have an amount of force that connects the dominant hand and the object that is the gun as both the hands are linked and

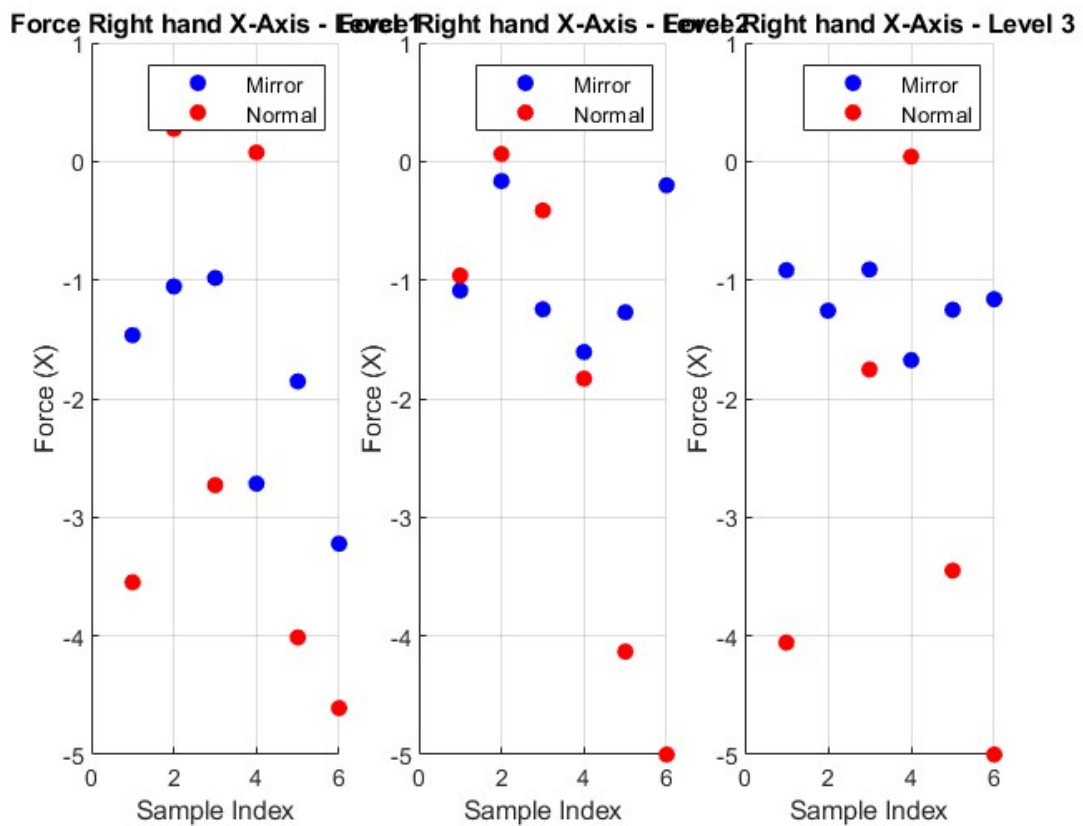


Figure 4.11 Forces of Right hand X-axis. Here the X-axis represents the sample index. This likely refers to the order in which the data points were collected during the experiment, the Y-axis represents the force value in Newtons (N). and the two sets of data points are plotted in each subplot using different colors (blue and red). These represent forces measured in two different modalities.

fixed to it. There is an individual link between the two which we can switch hands as we can regulate the force that the other hand needs to the move object in contact. If it does not move and you move in the other direction, and the non-dominant hand does not follow the link will be broken and we lose the point. This is a specific exercise for robotic mirror therapy. The force is kept to 5 newtons but it could go up to 15 newtons keeping the safety of the device which is a safe space for the experiment.

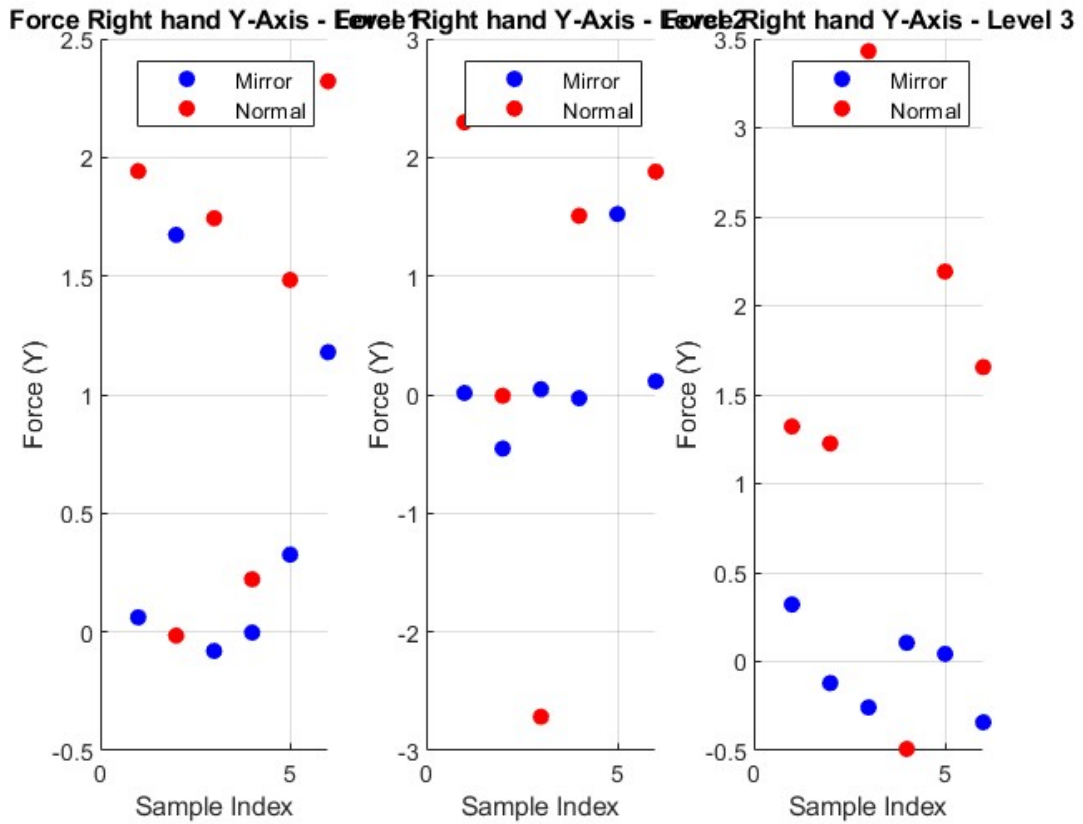


Figure 4.12 Forces of Right hand Y-axis. Here the X-axis represents the sample index. This likely refers to the order in which the data points were collected during the experiment, the Y-axis represents the force value in Newtons (N). and the two sets of data points are plotted in each subplot using different colors (blue and red). These represent forces measured in two different modalities.

4.6 Conclusion

I have worked on the integration of the game and virtual environment with the Exergame and my Contribution is to create different levels and aspects of graphical unity and C# Controlling the Alex. I worked together on the control part but most of my contribution was in the aspect of Exergamegame and the 3 levels. Then I did the Experiments designing the protocol and finally the data analysis to see the performance of both modalities. I got these results and what was the meaning of these results was the performance was better in the mirror modality compared to the normal modality. It is better for rehabilitation or exergaming, and we get more feedback that is realistic and that can help you in the movements thus I can conclude by answering the research questions that the mirror modality helps in improved performance

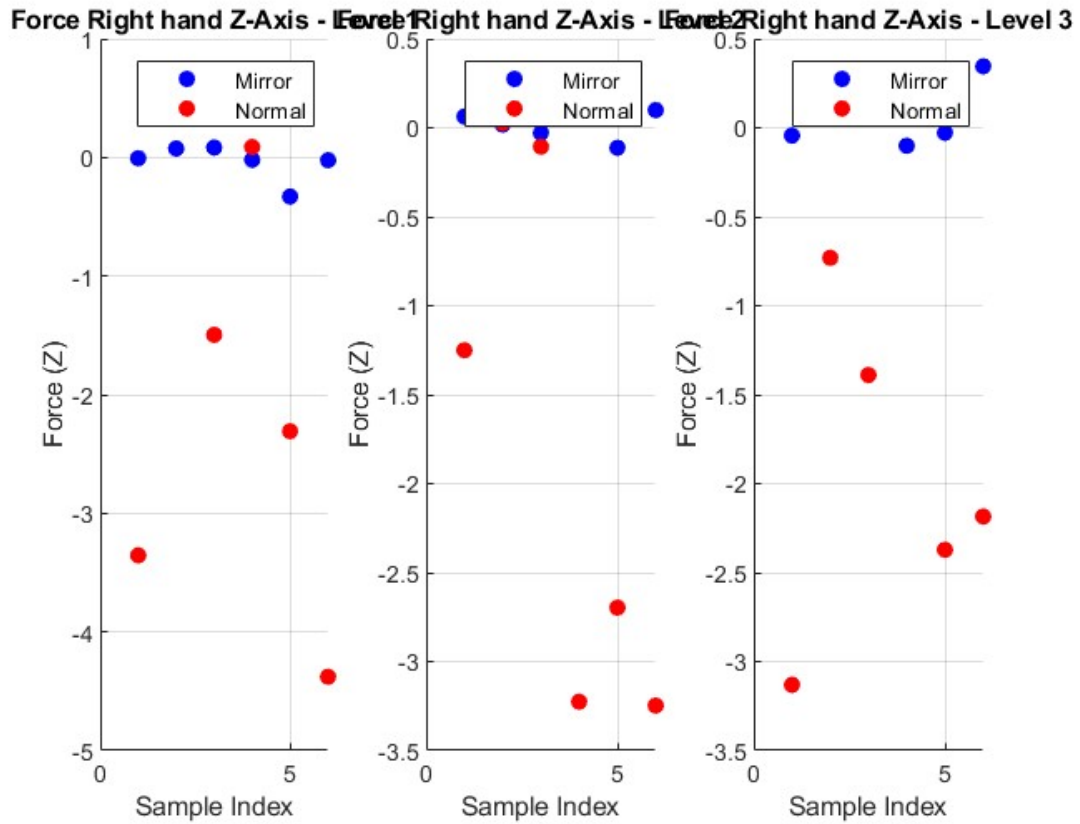


Figure 4.13 Forces of Right hand Z-axis. Here the X-axis represents the sample index. This likely refers to the order in which the data points were collected during the experiment, the Y-axis represents the force value in Newtons (N). and the two sets of data points are plotted in each subplot using different colors (blue and red). These represent forces measured in two different modalities.

compared to the Normal modality which is the haptic feedback modality and also the mirror modality is beneficial for promoting the motor function in the affected hand in the beginning stages.

Future Improvements

Change is a part of life and for the betterment of the protection of human beings and the environment, the thinking and practices should be changed as the new tests and improvements are the key to it as it is essential to take advantage of experience from the previous studies, as it reveals that prevalidation makes formal validation studies faster, less expensive, and more likely to be successful (21). As we have implemented the Robotic Mirror modality there could be more possibilities to make this study improve further and the improvements can be

made or updated as the years pass by so as per my observation, the following improvements can be made for this project in the future.

1. **Studies with motor impaired patients** can be used and further developed.
2. **Additional quantification and comparisons** can be made.
3. **Optimized therapies based on evidence** as it will enhance the study more adequately.

These are a few of many improvements that can be made but the main question is that with every study there is a conclusion that can be the turning point or the beginning of some innovations the world might need.

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Appendix A

Questioner for participants

Below is the questionnaire for the participants who took this exercise for the data analysis. the information is collected and organized with respect to this project and is gathered from previous studies conducted in Arise lab the Questions are based on the experience of the participants with the basic knowledge of Robotic Mirror Therapy and Rehabilitation.

Questionnaire

Do you currently have any neurological conditions or diseases?

☐ Yes

☐ No

Are you sick today?

☐ Yes

☐ No

1. Questions

Directions: Please indicate your level of agreement or disagreement with each of these statements. Place an "X" mark in the box of your answer.

	Strongly Agree	Agree	Neutral	Disagree	Strongly Disagree
1. I am familiar with Robotic Mirror Therapy.					
2. I have Experience with Virtual Reality.					
3. I have used the Exoskeleton for Rehabilitation purpose before.					
4. I am Satisfied with my current Upper Limb function.					
5. I am very curious about Stroke Rehabilitation Techniques					
6. I play Video games regularly.					

Figure A.1 Questioner 1

1. Questions

Directions: Please indicate your level of agreement or disagreement with each of these statements regarding Haptic Feedback Modality. Place an "X" mark in the box of your answer.

	Strongly Agree	Agree	Neutral	Disagree	Strongly Disagree
1. I enjoyed doing the exercise.					
2. The exercise was too difficult to perform.					
3. I was frustrated with the exercise.					
4. I was comfortable using the VR Headset.					
5. dizziness.					
6. Headache.					
7. Eyestrain.					
8. I would like to do the exercise again.					

Figure A.2 Questioner 2

2. Questions

Directions: Please indicate your level of agreement or disagreement with each of these statements regarding Mirror Haptic Feedback Modality. Place an "X" mark in the box of your answer.

	Strongly Agree	Agree	Neutral	Disagree	Strongly Disagree
1. I enjoyed doing the exercise.					
2. The exercise was too difficult to perform.					
3. I was frustrated with the exercise.					
4. I was comfortable using the VR Headset.					
5. dizziness.					
6. Headache.					
7. Eyestrain.					
8. I would like to do the exercise again.					

Figure A.3 Questioner 3

How would you rate your overall experience in the Haptic Feedback Modality?

- ☐ **Highly satisfactory**
- ☐ **Satisfactory**
- ☐ **Neutral**
- ☐ **Unsatisfactory**
- ☐ **Highly Unsatisfactory**

How would you rate your overall experience in the Robotic Haptic Feedback Modality?

- ☐ **Highly satisfactory**
- ☐ **Satisfactory**
- ☐ **Neutral**
- ☐ **Unsatisfactory**
- ☐ **Highly Unsatisfactory**

Which Modality do you find more enjoyable to use?

- ☐ **Haptic Feedback Modality**
- ☐ **Mirror Haptic Feedback Modality**
- ☐ **No difference**

Why did you choose this modality as more enjoyable?

Figure A.4 Questioner 4

Appendix B

Alex RS Technical Data sheet

The below sheet represents all the technical data that is important and related to the exoskeleton used in this thesis and is taken from the official website <http://www.wearable-robotics.com/kinetek/>



Technical data sheet - ALEx RS



GENERAL INFORMATION:

Full name: Arm Light Exoskeleton Rehab Station rev. 3.1

Classification: Class IIa Medical Device (Annex 9, Rule 9 Dir. 93/42/EEC - MDD)

Intended use: Motor rehabilitation of upper limbs in neurological or orthopedic field

FEATURES:

Used technologies: Exoskeletal Robotics, *Serious Games*

Device dimensions: 2124 mm x 3290 mm x 2120 mm

Weight: ~160 kg

Power supply: 100-230 VAC, 50/60 Hz, 1000 VA

Robotic exoskeletons number: 2

Moving parts weight of the exoskeletons: 5,6 kg

Exoskeletons Degrees of Freedom: 6 (4 actuated and sensorized + 2 sensorized)

Sensorized exoskeletons handles: Yes

Integrated application display Unit: Yes

Application display Unit dimensions: 43"

Operator Monitor dimensions: 23,8 "

Adjustable seat height: Yes

Adjustable exoskeleton distance: Yes

Possibility to use the exoskeletons individually or simultaneously: Yes

Permitted patient characteristics: min. height 1.50 m, max. height 1.90 m, weight $\leq$ 130 kg

Direct use of exoskeletons from own wheelchair: No

Openable and adjustable footrest: Yes

Wheels for moving the device: Yes

Database for patient management: Yes

Patient capacity evaluation functions: Yes

Automatic reports to monitor therapy progress: Yes

TECHNICAL SPECIFICATIONS:

Mobility fields for each DoF: Shoulder Adduction/Abduction [0° ; +110°]

External/Internal Shoulder Rotation [-60° ; +40°]

Shoulder Flexion/Extension [-10° ; +155°]

Elbow Flexion/Extension [-90° ; +70°]

Forearm Prono-Supination [-80° ; +80°]

Wrist Flexion/Extension [-50° ; +50°]

MAX. peak force at End Effector (each direction with extended arm): 70 N

MAX. continuous force at the End Effector (each direction with extended arm): 50 N

MAX. peak single torque for each articular joint: Shoulder Adduction/Abduction: 45 Nm

Shoulder Rotation: 45 Nm

Shoulder Flexion/Extension: 35 Nm

Elbow Flexion/Extension: 30 Nm

MAX. continuous single torque for each articular joint: Shoulder Adduction/Abduction: 35 Nm

Shoulder Rotation: 35 Nm

Shoulder Flexion/Extension: 25 Nm

Elbow Flexion/Extension: 20 Nm

MAX. joint speed (single movements, nil torques): 200 °/s (all joints)

End Effector position accuracy: 20 mm

End Effector position resolution: 0.05 mm

Joint position accuracy: 2°

Joint position resolution: 0.005°

End Effector force accuracy: 8 N

End Effector force Resolution: 0.2 N

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Technical data sheet
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